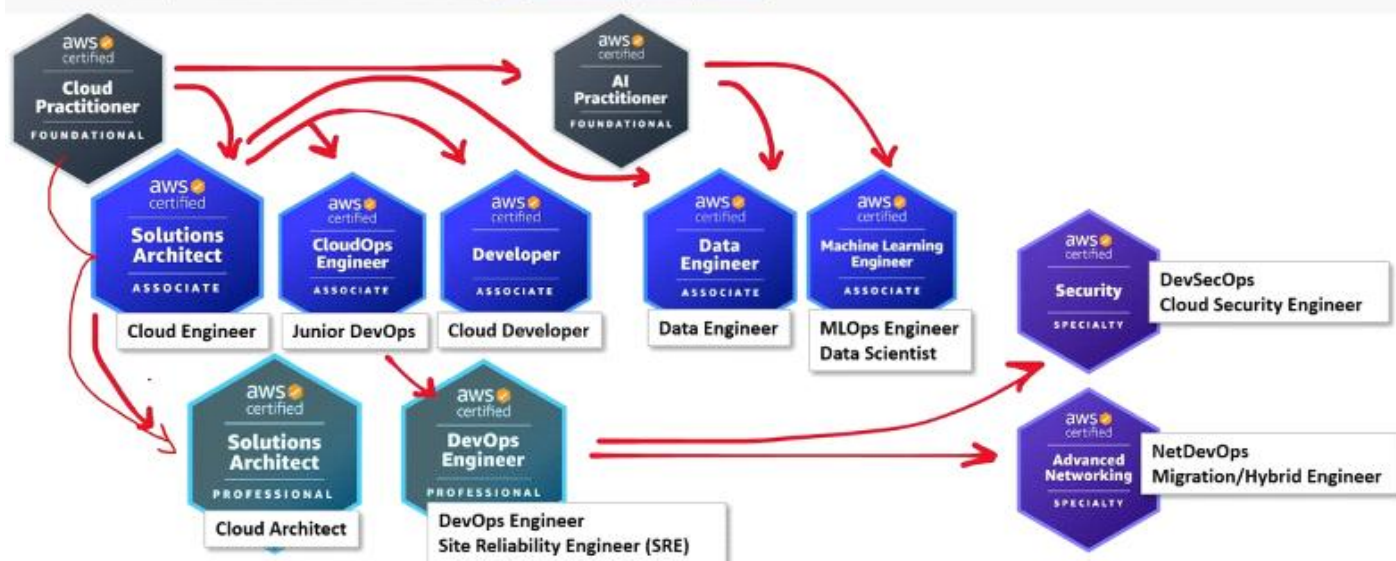


The AWS Certification Roadmap

Cheat sheets, Practice Exams and Flash cards www.exampmro.co/clf-c02



➤ **Questions will be ask within 4 Domain :**

Domain	Weight	Key Topics
Cloud Technology and Services	34%	Core AWS services, compute, storage, DB, networking
Security and Compliance	30%	Shared responsibility model, IAM, compliance
Cloud Concepts	24%	Value of cloud, global infrastructure, Well-Architected
Billing, Pricing and Support	12%	Pricing models, Cost Explorer, support plans

Cloud Computing

Cloud computing is the delivery of computing services—such as servers, storage, databases, networking, software, and analytics—over the internet instead of using your own local computer or physical servers.

On-Premise

- You own the servers
- You hire the IT people
- You pay or rent the real-estate
- You take all the risk

Cloud Providers

- Someone else owns the servers
- Someone else hires the IT people
- Someone else pays or rents the real-estate
- You are responsible for your configuring cloud services an code. someone else takes care of the rest.

❖ AWS was launched in 2006 as a leading cloud service provider.

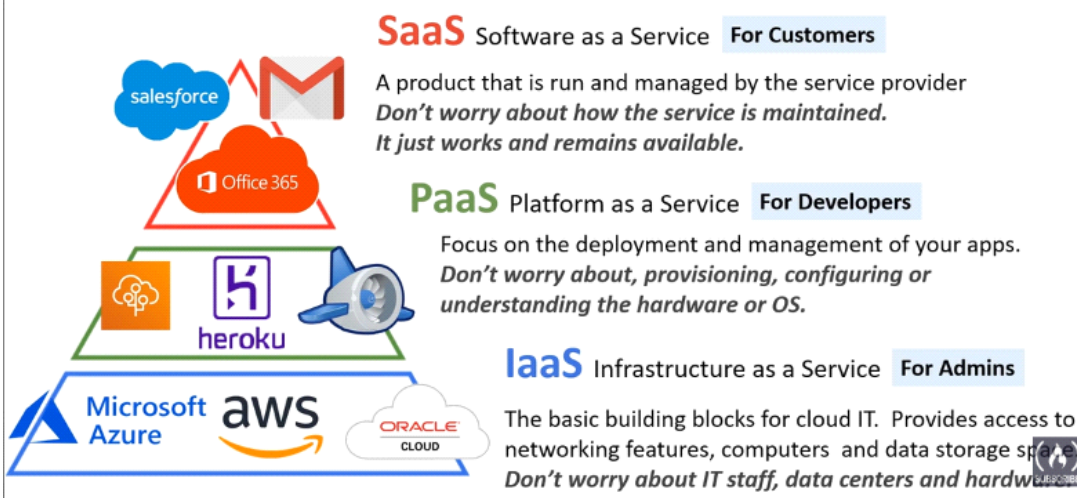
➤ **These are the cloud service Provider :**



Cloud Service Provider (CSPs) that are **Infrastructure as a Service (IaaS)** will always have **4 core cloud service** offerings:



➤ **Types of Cloud Computing :**



➤ **Types of deployment Model :**

- Public Cloud** : Everything (workload or project) is built on the CSP. Also known as "Cloud Native or Cloud first"
- Private cloud** : Everything built on the company's datacenters. Also known as on-premise.
- Hybrid cloud** : Using both on-premises and cloud service provider.
- Cross - cloud** : Using multiple cloud providers. **AKA multi-cloud**

➤ **Deployment Model use case :**



Benefits of cloud :



The benefits of the cloud is a summary of reasons why an organization would consider adopting or migrating to utilizing public cloud.

- **Agility**
 - Increase speed and agility
- **Pay-as-you go pricing**
 - Trade capital expense for variable expense
- **Economy of scale**
 - Benefit from massive economies of scale
- **Global Reach**
 - Go global in minutes
- **Security**
- **Reliability**
 - Stop spending money on running and maintaining data centers
- **High Availability**
- **Scalability**
 - Benefit from massive economies of scale
- **Elasticity**

Regions are **geographically distinct locations** consisting of one or more Availability Zones.



Every region is **physically isolated** from and independent of every other region in terms of **location, power, water supply**

Selecting a Region

Consider these factors when choosing a Region:

Compliance

Some regulations require data to stay within specific countries. For example, GDPR requires companies operating in the EU to choose Regions that meet its data protection requirements.

Customer Proximity

Selecting a Region near your customers reduces latency and improves their experience. For example, a Seattle-based company with Norwegian customers might run infrastructure locally but deploy customer-facing applications in a European Region.

Feature Availability

Not all Regions support all services. AWS constantly expands, but some features may only be available in specific Regions. For example, AWS GovCloud is only available in Regions designed for US government requirements.

Pricing

Costs vary by Region based on operational expenses, taxes, and local requirements. You will learn more about pricing in a later chapter.

What are AWS Edge Locations?

AWS Edge Locations are strategically placed sites around the world that cache content to deliver data with lower latency. They are smaller facilities outside of standard Regions, positioned closer to end users.



An American multinational computer technology corporation headquartered in **Seattle, Washington**



Amazon was founded in 1994 by **Jeff Bezos** and the company started as an online store for books and expanded to other products.



@timothyeberly on Unsplash 

Which of the following components of the AWS Global Infrastructure consists of one or more discrete data centers interconnected through low latency links?

- A. Availability Zone
- B. Edge location
- C. Region
- D. Private networking

What is an example of agility in the AWS Cloud?

- A. Access to multiple instance types
- B. Access to managed services
- C. Using Consolidated Billing to produce one bill
- D. Decreased acquisition time for new compute resources**

Cloud में Agility = जरूरत के अनुसार जल्दी से resources लेना, बदलना या हटाना।

Which feature of the AWS Cloud will support an international company's requirement for low latency to all of its customers?

- A. Fault tolerance
- B. Global reach**
- C. Pay-as-you-go pricing
- D. High availability

Which of the following security-related services does AWS offer? (Choose two.)

- A. Multi-factor authentication physical tokens
- B. AWS Trusted Advisor security checks**
- C. Data encryption**
- D. Automated penetration testing
- E. Amazon S3 copyrighted content detection

Which of the following is a benefit of using the AWS Cloud?

- A. Permissive security removes the administrative burden.
- B. Ability to focus on revenue-generating activities.**
- C. Control over cloud network hardware.
- D. Choice of specific cloud hardware vendors.

Which option is an advantage of AWS Cloud computing that minimizes variable costs?

- ☐ High Availability Reduces downtime but doesn't directly lower costs.
- ☒ Economics of Scale AWS's massive scale = **lower pay-per-use rates** for you (e.g., S3 storage costs drop over time due to economies of scale).
- ☐ Global Reach Enables multi-region deployment but isn't cost-focused.
- ☐ Agility Speeds up innovation but isn't tied to cost optimization.

[Documentation](#) > [AWS Whitepapers](#) > [AWS Whitepaper](#)

Six advantages of cloud computing

[PDF](#)

[RSS](#)

☐ Focus mode

A company wants to build graph queries for real-time fraud pattern detection.

Which AWS service will meet this requirement?

- ☒ Amazon Neptune Amazon Neptune is a fully managed graph database designed to identify complex relationships between entities like users, transactions, or devices.
- ☐ Amazon DynamoDB NoSQL key-value/wide-column store, not optimized for graph traversals.
- ☐ Amazon Timestream Time-series database (e.g., IoT telemetry), unrelated to fraud graphs.
- ☐ Amazon Forecast ML-based time-series forecasting, not graph analysis.

An online retail company wants to integrate its online shopping website with social media login credentials.

Which AWS service can the company use to make this integration?

- ☐ AWS Directory Service For Microsoft AD or LDAP (not social logins).
- ☐ AWS Identity and Access Management (IAM) Manages AWS resource access (not end-user social auth).
- ☐ Amazon Cognito Amazon Cognito is designed for federated identity and social media login integration (e.g., Facebook, Google, Apple). User Pools, Identity Pools
- ☐ AWS IAM Identity Center (AWS Single Sign-On)

AWS Shared Responsibility Model

Security and Compliance is a shared responsibility between AWS and the customer.

AWS is responsible for: **Security of the cloud** - protecting the infrastructure (hardware, software, hypervisor, networking, facilities).

You are responsible for: **Security in the cloud** - managing your data, access controls, and encryption.

➤ **Types of Controls**

1. Inherited Controls (AWS Only)

- Physical security
- Environmental controls (power, cooling, etc.)

2. Shared Controls (AWS + Customer)

Control	AWS	Customer
Patch Management	Infrastructure patches	OS & application patches
Configuration Management	AWS infrastructure	EC2, databases, applications
Awareness & Training	AWS employees	Customer employees

3. Customer-Specific Controls

Only the customer is responsible.

Examples:

- Data classification
- Data encryption
- IAM permissions
- Network security (VPC, Security Groups, NACLs)
- Application security

Authentication vs Authorization

Authentication = Who are you? (Verifies identity through credentials)

Authorization = What are you allowed to do? (Determines which actions a verified user can perform)

Under the shared responsibility model, which of the following tasks are the responsibility of the AWS customer? (Choose two.)

- A. Ensuring that application data is encrypted at rest**
- B. Ensuring that AWS NTP servers are set to the correct time
- C. Ensuring that users have received security training in the use of AWS services**
- D. Ensuring that access to data centers is restricted
- E. Ensuring that hardware is disposed of properly

Which of the following is a shared control between the customer and AWS?

- A. Providing a key for Amazon S3 client-side encryption
- B. Configuration of an Amazon EC2 instance
- C. Environmental controls of physical AWS data centers
- D. Awareness and training**

For which auditing process does AWS have sole responsibility?

- A. AWS IAM policies
- B. Physical security**
- C. Amazon S3 bucket policies
- D. AWS CloudTrail Logs

Which of the following is the customer's responsibility under the AWS shared responsibility model?

- A. Patching underlying infrastructure
- B. Physical security
- C. Patching Amazon EC2 instances**
- D. Patching network infrastructure

Under the shared responsibility model, which of the following is a shared control between a customer and AWS?

- A. Physical controls
- B. Patch management**
- C. Zone security
- D. Data center auditing

Which task is AWS responsible for in the shared responsibility model for security and compliance?

- A. Granting access to individuals and services
- B. Encrypting data in transit
- C. Updating Amazon EC2 host firmware**
- D. Updating operating systems

Which of the following services falls under the responsibility of the customer to maintain operating system configuration, security patching, and networking?

- A. Amazon RDS
- B. Amazon EC2**
- C. Amazon ElastiCache
- D. AWS Fargate

Which of the following is a component of the shared responsibility model managed entirely by AWS?

- A. Patching operating system software
- B. Encrypting data
- C. Enforcing multi-factor authentication
- D. Auditing physical data center assets**

IAM

Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources. With IAM, you can manage permissions that control which AWS resources users can access.

By default, all actions are denied - you must explicitly grant permission.

IAM Identities

Root User

The account owner created when you first start an AWS account with full permissions. Best practices:

- Use a strong password and enable MFA
- Avoid using root user for daily tasks
- Create IAM identities for daily work

IAM Users

Represents a person or application that interacts with AWS. Created without permissions by default - create one IAM user per individual.

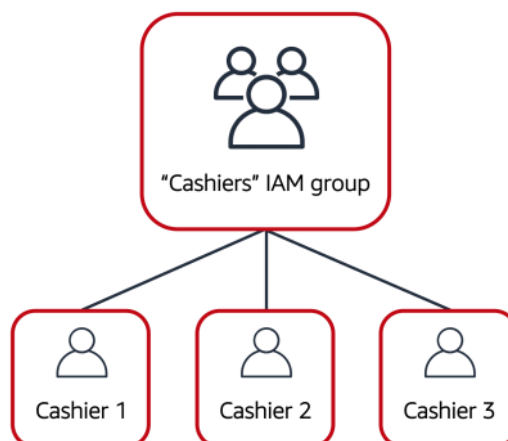
IAM Policies

Documents that allow or deny permissions to AWS resources. Customize access for each user based on what they need.

```
{
  "Version": "2012-10-17",
  "Statement": {
    "Effect": "Allow",
    "Action": ["s3:ListObject", "s3:GetObject"],
    "Resource": ["arn:aws:s3::: awsdoc-example-bucket",
      "arn:aws:s3::: awsdoc-example-bucket/*"]
  }
}
```

IAM Groups

A collection of IAM users. Policies assigned to a group apply to all users in that group.



IAM Roles

An identity you can assume to gain temporary access to permissions. When assuming a role, you abandon previous permissions

and take on the role's permissions. Best for temporary access scenarios.

Additional Services

Service	What It Does
IAM Identity Center	Centralizes identity management and single sign-on across AWS accounts
AWS Secrets Manager	Securely manage and rotate credentials, API keys, and secrets
AWS Systems Manager	Centralized view of nodes with automated patching and management

To understand :

1. IAM identity center - [Day 143 | AWS IAM Identity Center | What Is AWS IAM Identity Center ? | AWS In Hindi](#)

IAM	Easy meaning	Used by
User	An individual identity with AWS access	Person/application
Group	Collection of IAM users	Multiple users
Role	Temporary permissions that can be assumed	AWS services, users, applications

User details

User name

andrewbrown2

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ Provide user access to the AWS Management Console - optional

In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☒ Autogenerated password

You can view the password after you create the user.

☐ Custom password

Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } ' "

☐ Show password

☒ Users must create a new password at next sign-in - Recommended

Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

! If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Global Services :

- i. IAM
- ii. AWS Billing & Cost Management
- iii. Route 53
- iv. Amazon CloudFront

Regional Services :

- i. EC2
- ii. VPC
- iii. RDS
- iv. DynamoDB
- v. Lambda
- vi. ECS
- vii. EKS
- viii. CloudWatch
- ix. SNS
- x. Functions
- xi. Redshift
- xii. Glue
- xiii. EMR
- xiv. Athena
- xv. Kinesis

NOTE : Us-East-1 has most availability zone i.e 6

Which of the following Identity and Access Management (IAM) entities is associated with an access key ID and secret access key when using AWS Command Line Interface (AWS CLI)?

- A. IAM group
- B. IAM user**
- C. IAM role
- D. IAM policy

An organization wants to allow its employees to access AWS resources across multiple AWS accounts without sharing credentials.

Which AWS feature can facilitate this by providing temporary credentials?

- ☒ AWS Organizations **Helps manage multiple AWS accounts under a single organization for consolidated billing and policy management.**
- ☒ AWS SCT **Schema Conversion Tool (SCT) is used for heterogeneous database migration and schema conversion.**
- ☒ IAM Cross-account Roles **Allows users from one AWS account to access resources in another AWS account by assuming a role that provides temporary credentials.**
- ☐ Amazon Cognito **Provides authentication, authorization, and user management for web and mobile apps. It is **primarily designed for end-user authentication** rather than cross-account access**

A company has teams that have different job roles and responsibilities. The company's employees often change teams. The company needs to manage permissions for the employees so that the permissions are appropriate for the job responsibilities.

Which IAM resource should the company use to meet this requirement with the LEAST operational overhead?

- ☒ IAM user groups Requires manual user reassignment between groups when teams change (higher overhead).
- ☒ IAM roles IAM roles allow you to assign permissions based on job functions, not individual users. **No policy updates needed, Temporary credentials, Cross-account/team flexibility**
- ☒ IAM instance profiles For EC2 instances (not human users).
- ☐ IAM policies for individual users Hard to manage at scale (high overhead).

A developer wants AWS users to access AWS services by using temporary security credentials.

Which AWS service or feature should the developer use to provide these credentials?

- ☒ IAM policies Define permissions (don't issue credentials).
- ☒ IAM user group Manage long-term users (not temporary credentials).
- ☒ AWS Security Token Service (AWS STS) AWS STS generates temporary security credentials (access keys, session tokens) for IAM roles, Federated users.
- ☐ AWS IAM Identity Center (AWS Single Sign-On) Centralizes access but uses STS under the hood for temp credentials.

What is Amazon EC2?

Amazon EC2 (Elastic Compute Cloud) provides virtual servers in the AWS Cloud. Get computing power on-demand without buying physical hardware, and scale capacity as needed.

Working with EC2 involves three steps:

1. Launch

Select a template with your operating system and applications, choose an instance type (hardware configuration), and specify security settings to control network traffic.

2. Connect

Connect to your instance through various methods: remote desktop login, SSH, or programmatic access for applications.

3. Use

Run commands, install software, add storage, and manage files just like on a physical server.

Instance Type Categories

<u>Instance Type</u>	<u>Optimized For</u>	<u>Example Use Case</u>
General Purpose (T, M)	Balanced CPU & Memory	Web Applications
Compute Optimized (C)	CPU	Gaming, Batch Processing
Memory Optimized (R)	RAM	Databases, Caching
Storage Optimized (I, D, H)	Fast Storage	Big Data, NoSQL
Accelerated Computing (P, G)	GPU	AI/ML, Graphics
HPC	High Computing Power	Scientific Simulations

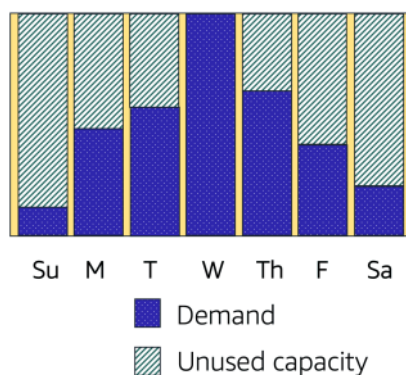
EC2 Auto Scaling

When servers receive more requests than they can handle, it can cause timeouts and outages. EC2 Auto Scaling automatically adds or removes instances to match demand.

Two Approaches to EC2 Auto Scaling

- **Dynamic scaling:** responds to changing demand in real-time
- **Predictive scaling:** schedules instances based on predicted demand

You can combine both approaches for faster scaling.



Auto Scaling Groups

An Auto Scaling group is a collection of EC2 instances. You define capacity limits:

- **Minimum:** Always keep at least this many instances running
- **Desired:** Ideal number of instances for normal traffic
- **Maximum:** Never exceed this many instances (controls costs)

Example: Set min=2, desired=2, max=10. Normally 2 instances run. During traffic spikes, Auto Scaling adds up to 10, then scales back down when traffic drops.

Elastic Load Balancing (ELB)

Elastic Load Balancing (ELB) distributes incoming traffic across multiple EC2 instances. It acts as a single point of contact, spreading requests evenly so no single instance gets overloaded.

ELB works seamlessly with **EC2 Auto Scaling** to enhance performance and ensure high availability. It scales automatically with traffic at no additional hourly cost.

ELB Features

Key Benefits

- **Efficient distribution:** Evenly spreads traffic, preventing overload on any single instance
- **Automatic scaling:** Adjusts to traffic changes automatically
- **Simplified management:** Handles maintenance, updates, and failover

Routing Methods (how we distribute traffic between instances)

- **Round Robin:** Distributes traffic evenly in a cyclic manner
- **Least Connections:** Routes to the server with fewest active connections
- **IP Hash:** Uses client IP to consistently route to the same server
- **Least Response Time:** Directs to the server with fastest response

Pricing Models : Important topic

1. **Reserved Instances** offer up to 72% discount over On-Demand for 1-year or 3-year commitments — perfect for predictable, steady-state workloads
NOTE : Reserved Instance → Don't need it anymore → Eligible? → **Reserved Instance Marketplace** → Sell remaining term
2. **Spot Instances** are cheaper (up to 90% off) but can be interrupted by AWS — **unsuitable for continuous 24/7 applications**
3. **On-Demand** is most expensive (pay per hour).
4. **Dedicated Hosts** provide dedicated physical servers (for compliance), not a cost savings model.

- **On-Demand**
 - No commitment.
 - Highest cost per hour.
 - Best for unpredictable workloads.
- **Reserved Instances**
 - Commit for 1 or 3 years.
 - Lower cost than On-Demand.
 - Best for steady-state applications.
- **Spot Instances**
 - Lowest cost.
 - Uses unused AWS capacity.
 - Price varies with supply and demand.
 - Can be interrupted by AWS.
- **Convertible Reserved Instances**
 - Long-term commitment.
 - Can change instance attributes.
 - More flexible than Standard Reserved Instances.

- Slightly less discount than Standard Reserved Instances.

NOTE : **Instance level firewalls** : Security group (virtual firewall that controls incoming and outgoing traffic)

Question 2 — AWS Pricing Models

A startup runs a predictable web application that requires a specific EC2 instance type 24/7 for the next 2 years. Which EC2 pricing model provides the MOST cost savings?

- A. On-Demand Instances
- B. Spot Instances
- C. Dedicated Hosts
- D. Reserved Instances (1 or 3-year)

Which Amazon EC2 pricing model adjusts based on supply and demand of EC2 instances?

- A. On-Demand Instances
- B. Reserved Instances
- C. **Spot Instances**
- D. Convertible Reserved Instances

Which of the following AWS features enables a user to launch a pre-configured Amazon Elastic Compute Cloud (Amazon EC2) instance?

- A. Amazon Elastic Block Store (Amazon EBS)
- B. **Amazon Machine Image**
- C. Amazon EC2 Systems Manager
- D. Amazon AppStream 2.0

A company is migrating an application that is running non-interruptible workloads for a three-year time frame.

Which pricing construct would provide the MOST cost-effective solution?

- A. Amazon EC2 Spot Instances
- B. Amazon EC2 Dedicated Instances
- C. Amazon EC2 On-Demand Instances
- D. Amazon EC2 Reserved Instances**

Web servers running on Amazon EC2 access a legacy application running in a corporate data center. What term would describe this model?

- A. Cloud-native
- B. Partner network
- C. Hybrid architecture**
- D. Infrastructure as a service

An organization wants to ensure that its Amazon EC2 instances are securely accessed over the internet.

Which AWS service allows the company to create instance level firewall rules to control inbound and outbound traffic to EC2?

- ☒ AWS Web Application Firewall (WAF) Designed to protect web applications from common web exploits. Operates at the application layer. Does not provide instance-level.
- ☒ AWS Network Firewall Managed service that provides network-level protection for your VPCs. It is used to secure network traffic within your VPC.
- ☒ AWS CloudFront Content delivery network (CDN) service that speeds up the distribution of your web content. It does not provide firewall capabilities.
- ☐ Security Groups Act as virtual firewalls for your EC2 instances to control incoming and outgoing traffic. They allow you to define rules that specify the allowed protocols, ports, and IP addresses for inbound and outbound traffic.

A company purchased Amazon EC2 Standard Reserved Instances (RIs) for a workload in the AWS Cloud. The company needs to move part of the workload to an instance family that does not match the instance family of these Standard RIs.

How can the company take advantage of the Standard RIs that it no longer needs?

- ☒ Contact the AWS Support team, and ask the team to sell the Standard RIs AWS Support won't buy back RIs and they don't handle the selling of Reserved Instances
- ☒ Sell the Standard RIs on the Amazon EC2 Reserved Instance Marketplace This is the only official way to offload RIs you no longer need (AWS doesn't buy them back). Works if the RIs have at least 30 days remaining on their term.
- ☒ Sell the Standard RIs as a third-party seller on the AWS Marketplace AWS Marketplace is for software/SaaS products, not RIs.
- ☐ Convert the Standard RIs to Savings Plans You can't convert RIs to Savings Plans (they're separate pricing models).

A company needs to provision uninterruptible Amazon EC2 instances, when needed, and pay for compute capacity by the second?

Which EC2 instance purchasing option will meet these requirements?

- ☒ Reserved Instances Require long-term commitment (1-3 years) for discounts.
- ☒ Spot Instances Cheap but can be interrupted (not suitable for uninterruptible needs).
- ☒ On-Demand Instances On-Demand Instances are always available. On-Demand supports per-second billing granularity. No long-term commitment required.
- ☐ Dedicated Instances Run on isolated hardware but don't affect billing granularity (still On-Demand or Reserved).

A company has data lakes designed for high performance computing (HPC) workloads. Which Amazon EC2 instance type should the company use to meet these requirements?

- ☒ General purpose instances
General purpose (e.g., M5, T3) → Balanced CPU/memory, but not optimized for HPC.
- ☒ Compute optimized instances
Compute optimized instances (e.g., C5, C6i, C7g) are designed for high-performance computing (HPC) workloads.
- ☒ Memory optimized instances
Memory optimized (e.g., R5, X2gd) → Best for in-memory databases, not pure compute.
- ☐ Storage optimized instances
Storage optimized (e.g., I3, D2) → For high disk throughput, not compute-heavy workloads.

A customer runs an On-Demand Amazon Linux EC2 instance for 3 hours, 5 minutes, and 6 seconds.

For how much time will the customer be billed?

- ☐ 3 hours, 5 minutes
- ☐ 3 hours, 5 minutes, and 6 seconds
- ☒ 3 hours, 6 minutes
- ☐ 4 hours

Per-second billing removes the cost of unused compute time from your bill. This particularly helps workloads that run over irregular time periods. Amazon EC2 usage is billed in one-second increments, with a minimum of 60 seconds. The same is true for provisioned storage for Amazon Elastic Block Store (Amazon EBS) volumes. Per-second billing applies to all purchase options. It's available across all Regions and Availability Zones for these instances:

- Amazon Linux
- Windows
- Red Hat Enterprise Linux
- Ubuntu
- Ubuntu Pro

EC2 instances are billed in one-second increments, with a minimum of 60 seconds.

Total Runtime → **3 hours, 5 minutes, and 6 seconds.**

AWS rounds up to the nearest minute for billing purposes.

A company plans to run a compute-intensive workload that uses graphics processing units (GPUs).

Which Amazon EC2 instance type should the company use?

- ☒ Compute Optimized High CPU but **no GPUs** (good for HPC, batch processing) → (e.g., C6i)
- ☒ General purpose Balanced CPU/memory (not GPU-accelerated) → (e.g., M6i)
- ☒ Storage optimized High disk throughput (unrelated to GPUs) → (e.g., I4i)
- ☐ Accelerated computing (e.g., P4, G5, Inf1) are purpose-built for **GPU workloads**, Agentic and generative AI applications, video and image generation, speech recognition, and more

A company is migrating its on-premises server to an Amazon EC2 instance. The server must remain active at all times for the next 12 months. However, the organization expects to change the compute family due to the evolving nature of the workload.

Which EC2 pricing option would be the MOST cost-effective for this scenario?

- ☒ On-Demand Most expensive (no long-term discounts).
- ☒ Reserved Instances Require locking into specific instance families (unless using Convertible RIs, which offer less flexibility than Savings Plans).
- ☒ Savings Plan Savings Plans offer flexibility + cost savings (up to 72% discount vs. On-Demand) for long-term workloads where the instance family might change
- ☐ Spot Instances Cheap but can be interrupted (unsuitable for always-on workloads).

An online research company wants to run a simulation for 10 months without interruptions.

Which Amazon EC2 instance purchasing option will meet these requirements MOST cost-effectively?

- ☒ Spot Instances ~90% cheaper but can be **interrupted** (unsuitable for non-stop workloads).
- ☐ On-Demand Instances Most expensive (no long-term savings).
- ☒ Dedicated Hosts Physical server isolation (expensive, no RI discounts).
- ☐ Reserved Instances Reserved Instances (RIs) offer up to 75% discount vs. On-Demand pricing for long-term, uninterrupted workloads (like this 10-month simulation).
Capacity reservation 1-year terms (even 10 months is cheaper than On-Demand).

A company is migrating to the AWS Cloud and plans to run experimental workloads for 3 to 6 months on AWS.

Which pricing model will meet these requirements?

- ☒ Use Savings Plans for a 3-year term. Requires a **long-term commitment** (3 years). Not cost-effective for **short-term workloads**.
- ☒ Use Dedicated Hosts Provides **physical servers** for compliance or licensing needs. **Expensive** and unnecessary for **temporary, experimental use**.
- ☒ Use On-Demand Instances Ideal for short-term, **unpredictable**, or **experimental** workloads. Perfect for **3-6 month experimental projects**.
- ☐ Buy Reserved Instances. Requires **1- or 3-year commitment**. Best for **steady-state, long-term workloads**, not short-term testing.

AWS Cloud Messaging and Queuing

Application Architecture

There are two main approaches to connecting components in an application architecture:

Monolithic

All components are **tightly coupled** and run as a single unit. If one component fails, the entire application can go down.

Microservices

Components are **loosely coupled** and communicate through messaging. If one component fails, others continue working normally.

Messaging and Queuing

To handle communication in a **microservices** architecture, we use messaging and queuing to transmit data and fulfill requests. AWS messaging services allows components to communicate asynchronously, so failures don't cascade through the application.

AWS Messaging Services

- **Amazon EventBridge:** Serverless event routing between apps, AWS services, and third-party software
- **Amazon SNS:** Publish-subscribe service for sending messages to multiple subscribers
- **Amazon SQS:** Message queuing for reliable component communication

What is Amazon EventBridge?

EventBridge is a serverless service that connects application components using events. It routes events from custom apps, AWS services, and third-party software, simplifying how you receive, filter, transform, and deliver events.

EventBridge Features

Key Benefits

- **Automatic routing:** Route events to relevant services automatically
- **High volume:** Handle large event volumes during peak times
- **Independence:** Each service works independently
- **Reliability:** Stores events if a service fails and processes them when available

NOTE : A **serverless service** means **you don't have to manage or maintain the servers**.

What is Amazon SNS?

Amazon Simple Notification Service (SNS) is a fully managed messaging service for sending messages to multiple recipients (subscribers). It decouples distributed systems and microservices, and can send notifications to users via SMS, mobile push, or email.

SNS can publish messages to:

- HTTP/HTTPS
- Email
- Amazon SQS
- AWS Lambda
- Mobile applications
- SMS (region-dependent)

What is Amazon SQS?

Amazon Simple Queue Service (SQS) is a message queuing service for communicating between software components. It sends, stores, and receives messages at any scale, ensuring messages aren't lost even when services are temporarily unavailable.

How It Works

An application places messages into a queue. A user or service retrieves the message, processes it, and removes it from the queue. Subscribers can be web apps, servers, Lambda functions, and more.

When to Use Each Service

Use EventBridge when:

- You need to route events from multiple sources (AWS services, SaaS apps, custom apps)
- You want to filter and route events based on content
- You're building event-driven architectures

Use SNS when:

- You need to send one message to many subscribers
- You want to send notifications (email, SMS, mobile push)
- You need fan-out messaging to multiple endpoints

Use SQS when:

- You need reliable message delivery between two components
- You want to decouple sender and receiver
- Messages should be processed in order or only once

What is AWS Lambda?

AWS Lambda is a serverless compute service that runs code in response to events. No servers to manage, automatic scaling, and you pay only for compute time used (down to the millisecond).

How AWS Lambda Works

1. **Upload code:** Deploy your code as a Lambda function
2. **Set triggers:** Configure events to trigger your code (AWS services, HTTP requests, schedules)
3. **Code runs when triggered:** Lambda executes and scales automatically when triggered
4. **Pay only for compute time:** Charged based on execution time and memory used

Lambda Features

Use Cases

- **Image processing:** Resize images when uploaded
- **Real-time processing:** Handle streaming data or in-game events
- **API backends:** Run backend code for web and mobile apps

Supported Languages

Python, Node.js, Java, C#, Go, Ruby, plus custom runtimes for other languages.

A company wants to reduce the physical compute footprint that developers use to run code.
Which service would meet that need by enabling serverless architectures?

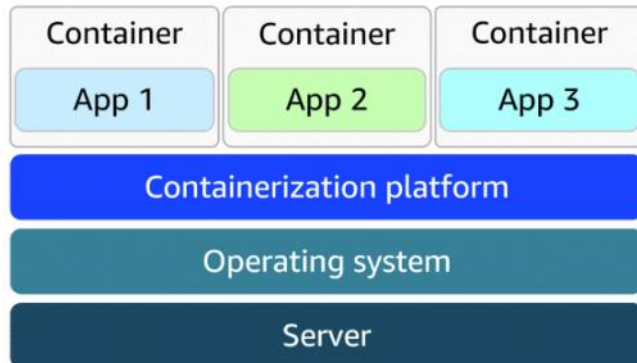
- A. Amazon Elastic Compute Cloud (Amazon EC2)
- B. AWS Lambda**
- C. Amazon DynamoDB
- D. AWS CodeCommit

What are Containers?

A container packages your application with everything it needs to run. Multiple containers can share the same server's operating system while keeping their applications and libraries isolated.

Container Benefits

Architecture



Multiple containers run on the same server, each with its own application, sharing the OS through a containerization platform

Deployment Consistency

Containers keep the application **environment identical** everywhere, making deployments smoother and debugging easier.

AWS Container Services

- **Orchestration:** [Amazon ECS](#) and [Amazon EKS](#) - deploy, scale, and manage containers
- **Registry:** Amazon ECR - store and version container images
- **Compute:** [AWS Fargate](#) - run containers without managing servers

What is Amazon ECS?

Amazon Elastic Container Service (ECS) is a container orchestration service for deploying, scaling, and managing containerized applications on AWS.

ECS Features

Launch Types

- **ECS with EC2:** Full control over infrastructure for custom hardware or networking needs
- **ECS with Fargate:** Serverless option - focus on code while AWS handles servers and scaling

Docker Support

ECS supports [Docker](#), a platform for building, testing, and deploying applications. AWS supports both open-source and enterprise Docker editions.

Learn more: [Amazon ECS](#)

What is Amazon EKS?

Amazon Elastic Kubernetes Service (EKS) is a fully managed service for running Kubernetes on AWS. It simplifies deploying, managing, and scaling containerized applications.

EKS Features

Kubernetes

Kubernetes is open-source software for container orchestration with a large community providing ongoing support. AWS keeps EKS updated with the latest Kubernetes features.

Launch Types

- **EKS with EC2:** Full control over infrastructure for complex, large-s

What is AWS Fargate?

AWS Fargate is a serverless compute engine for containers. No servers to provision or manage. Fargate handles the infrastructure so you can focus on developing your applications.

You pay only for the resources required to run your containers.

Fargate vs ECS/EKS

Fargate is a container **hosting platform**, while Amazon ECS and Amazon EKS are container **orchestration services**.

Elastic Beanstalk

Fully managed service that **deploys, manages, and scales web applications**. Upload your code and it handles infrastructure, scaling, and monitoring. Supports Java, .NET, Python, Node.js, Docker, and more.

Good for: Web apps, APIs, mobile backends, microservices.

AWS Batch

Fully managed batch computing that automatically schedules and scales compute resources based on job requirements.

Good for: Scientific computing, financial analysis, media transcoding, ML training.

Amazon Lightsail

Provides virtual servers, storage, and databases with **predictable monthly pricing**. Designed for users who want a simplified AWS experience.

Good for: Basic web apps, low-traffic sites, dev/test environments, blogs.

AWS Outposts

Extends AWS infrastructure to your on-premises data center for a consistent hybrid cloud experience.

Good for: Low-latency apps, remote data processing, regulatory compliance.

Amazon CloudFront

Amazon CloudFront is AWS's Content Delivery Network (CDN). It stores cached copies of website content, such as images, videos, CSS, and JavaScript, in Edge Locations that are close to users. When users request the content, it is delivered from the nearest Edge Location instead of the main server, resulting in faster loading times and lower latency.

CloudFront Use Cases

- **Streaming video:** Videos play smoothly without buffering, even during peak times
- **Ecommerce:** Product images and pages load quickly during busy shopping seasons
- **Mobile apps:** Map data and images delivered quickly for responsive experiences

Caching - How AWS Edge Locations Work

Caching stores copies of frequently accessed data closer to users, reducing the need to fetch data from the origin server every time.

1. A user requests content (e.g., an image or video)
2. If not cached, the content is fetched from the origin and stored at the Edge Location
3. Future requests for the same content are served from the nearest Edge Location

Content is delivered faster because it comes from the Edge Location nearest to the user, not the origin server.

Provisioning

Provisioning means setting up and configuring the AWS services.

There are three main ways to interact with AWS:

- Management Console
- Command Line Interface (CLI)
- Software Development Kits (SDKs)

Managed Provisioning Tools

In addition to manual provisioning, AWS offers two managed tools that simplify deployment: **AWS Elastic Beanstalk** and **AWS CloudFormation**.

AWS Elastic Beanstalk

Provide your code and configuration settings, and Elastic Beanstalk automatically handles deployment, capacity adjustment, load balancing, auto-scaling, and health monitoring.

Ideal for quickly deploying web applications without managing underlying infrastructure.

Elastic Beanstalk is a web infrastructure management service that handles deployment and scaling for web applications. It automatically manages setup, configuration, scaling, and provisioning of underlying AWS services.

Services managed by Elastic Beanstalk include: **EC2, S3, RDS, and DynamoDB**

AWS CloudFormation

CloudFormation is an Infrastructure as Code (IaC) service that automates resource provisioning. You define your **AWS resources in a template file**, and CloudFormation provisions and configures them for you

Which AWS service provides the ability to manage infrastructure as code?

- A. AWS CodePipeline
- B. AWS CodeDeploy
- C. AWS Direct Connect
- D. AWS CloudFormation**

A company wants to create templates that the company can reuse to deploy multiple AWS resources.

Which AWS service or feature can the company use to meet this requirement?

- ☒ AWS Marketplace **Pre-packaged third-party software / SaaS products (not for custom resource templates).**
- ☒ Amazon Machine Image (AMI) **Custom VM images for EC2 (limited to single instances, not multi-resource stacks).**
- ☒ AWS CloudFormation **Reusable templates (in JSON/YAML) to deploy and manage AWS resources consistently.**
- ☐ AWS OpsWorks **Configuration management (Chef/Puppet) for deployments, not templating.**

Key Networking Components

How you connect your resources depends on your setup and requirements. The main building blocks are:

- **Amazon VPC (Virtual Private Cloud):** A logically isolated section of AWS where you launch resources in a virtual network you define.
- **Subnets:** Sections of a VPC used to organize resources. Can be public (internet-facing) or private (internal only).

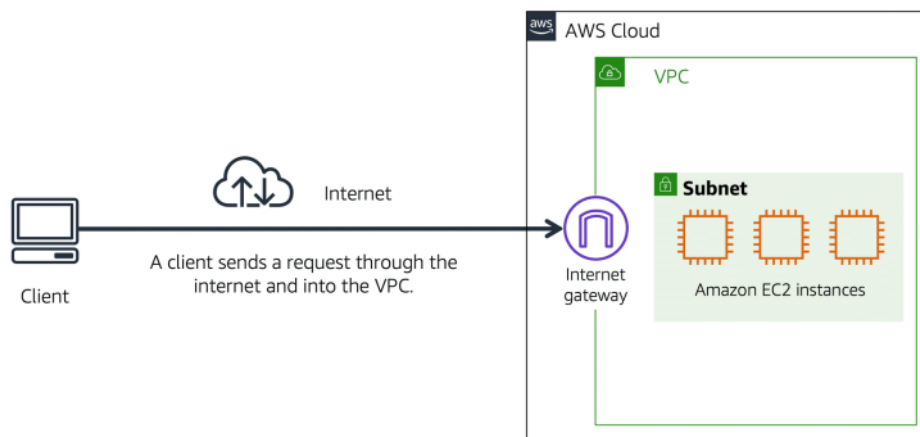
Amazon Virtual Private Cloud (VPC)

An Amazon Virtual Private Cloud (VPC) lets you isolate your AWS resources in a private network you define. You control which sections of the AWS Cloud to include and how to restrict network traffic.

Resources within a VPC can be organized into subnets, which are sections that contain specific resources.

Internet Gateway

An Internet Gateway connects your VPC to the public internet. Without it, resources in your VPC cannot be accessed from the internet.



Access Control

Network Access Control Lists (ACLs)

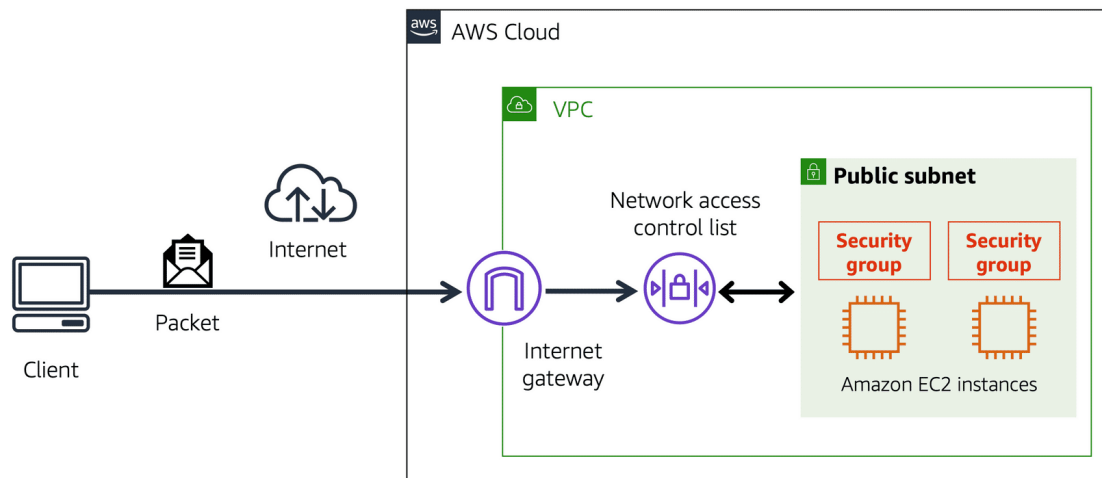
ACLs are firewalls that control inbound and outbound traffic at the subnet level. Packets on the approved list pass through; others are denied.

ACLs use **stateless packet filtering**: they check each packet against rules but have no memory of previous requests.

Security Groups

Security Groups are firewalls that control traffic for specific EC2 instances. By default, they deny all inbound traffic and allow all outbound. You add rules to customize this behavior.

Security Groups use **stateful packet filtering**: they remember previous actions and allow return traffic automatically.



Amazon Route 53

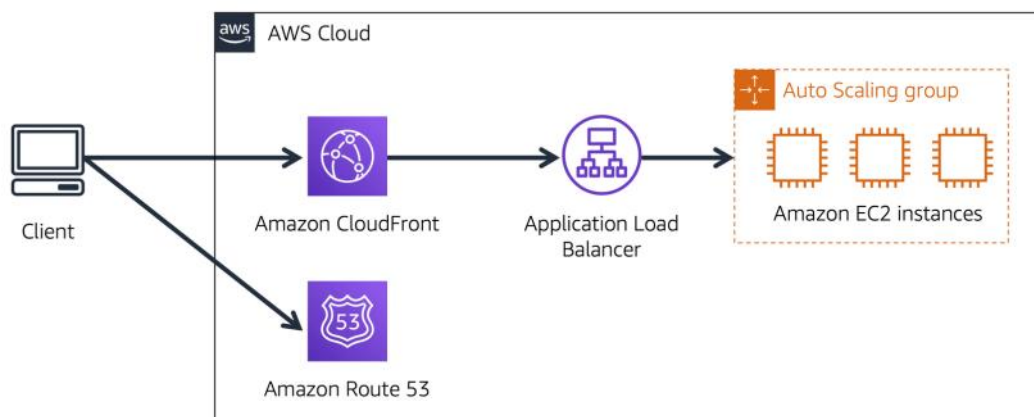
Amazon Route 53 is AWS's DNS service.

Its main job is to convert a **website name** into an **IP address**.

Route 53 is AWS's DNS web service that routes users to your applications running on EC2, load balancers, or even infrastructure outside AWS. **You can register new domains, transfer existing ones, and manage all DNS records in one place.**

Route 53 + CloudFront Architecture

Route 53 and CloudFront work together to deliver content efficiently:



1. User requests data from the application
2. Route 53 resolves the domain to an IP address
3. The request goes to the nearest CloudFront Edge Location
4. CloudFront connects to the Application Load Balancer
5. The Application Load Balancer forwards the request to an EC2 instance
6. A response is returned to the user

Edge Services Summary

- **Route 53:** Scalable cloud DNS service
- **CloudFront:** CDN for low-latency content delivery
- **Global Accelerator:** Routes traffic through AWS's private network for better performance

Global Architectures

AWS enables you to build applications that span multiple regions and data centers. Here are two common global architecture patterns:

- **Direct Connect Failover:** High-bandwidth dedicated connections with fault tolerance
- **Multi-Region Content Delivery:** Global content distribution using CloudFront and Route 53

Which of the following is an AWS managed Domain Name System (DNS) web service?

- A. Amazon Route 53
- B. Amazon Neptune
- C. Amazon SageMaker
- D. Amazon Lightsail

A company wants to connect its supported AWS services and VPCs. The company does not want to expose its internal traffic to the public internet.

Which AWS service will meet these requirements?

- ☒ AWS PrivateLink AWS PrivateLink provides private connectivity between VPCs, AWS services, and on-premises networks without exposing your traffic to the public internet.
- ☐ AWS Site-to-Site VPN AWS Site-to-Site VPN enables secure connections between your on-premises network and your AWS environment.
- ☐ Amazon Connect Cloud-based contact center service that enables businesses to provide customer service at scale.
- ☐ AWS Internet Gateway AWS Internet Gateway allows communication between your VPC and the internet. It is used for internet access.

Which of the following are features of network ACLs as they are used in the AWS Cloud? (Choose two.)

- ☒ They are stateless. Inbound/outbound rules are evaluated independently. You must explicitly allow return traffic
- ☐ They are stateful. Describes security groups (not NACLs).
- ☐ They evaluate all rules before allowing traffic. NACLs stop at the first match.
- ☒ They process rules in order, starting with the lowest numbered rule, when deciding whether to allow traffic. Rules are evaluated from lowest to highest numbered rule
Example: If rule #100 denies SSH and rule #200 allows it, SSH is denied.
- ☐ They operate at the instance level. NACLs operate at the subnet level (security groups are instance-level).

AWS Storage Types

Block Storage

Low-latency storage volumes that attach to EC2 instances like physical hard drives.

Examples of block storage:

- **EC2 Instance Store:** Temporary, high-performance storage
- **Amazon EBS:** Persistent block storage volumes

Object Storage

Manages data as objects with unlimited scalability for unstructured data like documents, images, and videos.

Examples of object storage:

- **Amazon S3:** Store and retrieve any amount of data

File Storage

Shared file systems accessible over networks by multiple users and applications simultaneously.

Examples of file storage:

- **Amazon EFS:** Scalable NFS file system
- **Amazon FSx:** Managed file storage for Windows, Lustre, and more

Shared Responsibility for Storage

- **Fully managed:** AWS handles everything. You only manage data, access, and configuration
- **Managed:** AWS handles infrastructure. You only manage backups and encryption
- **Unmanaged:** You handle everything. You manage data management, backups, and encryption

What are Instance Stores?

An Instance Store is block-level storage physically attached to the EC2 instance host computer. Data persists during reboots but is lost when the instance is stopped or terminated.

Best for: Temporary storage needs like buffers, caches, and scratch data. Not recommended for data that needs to persist.

Benefits of Instance Stores

- **Is included free:** Comes attached to many EC2 instance types at no additional cost
- **Has high performance:** Extremely low-latency storage directly attached to the host server

What is AWS EBS?

Amazon Elastic Block Store (EBS) provides persistent block-level storage volumes for EC2 instances. Unlike instance stores, EBS data persists even when the instance is stopped or terminated.

Common use cases are: database hosting, application backups, rapid deployment of dev environments.

EBS Benefits

- **Data persistence:** Data remains available even if the EC2 instance stops
- **Migration:** Move volumes between Availability Zones using snapshots
- **Flexibility:** Attach volumes to different instance types, modify size and performance on the fly
- **Disaster recovery:** Snapshots provide reliable backups across regions

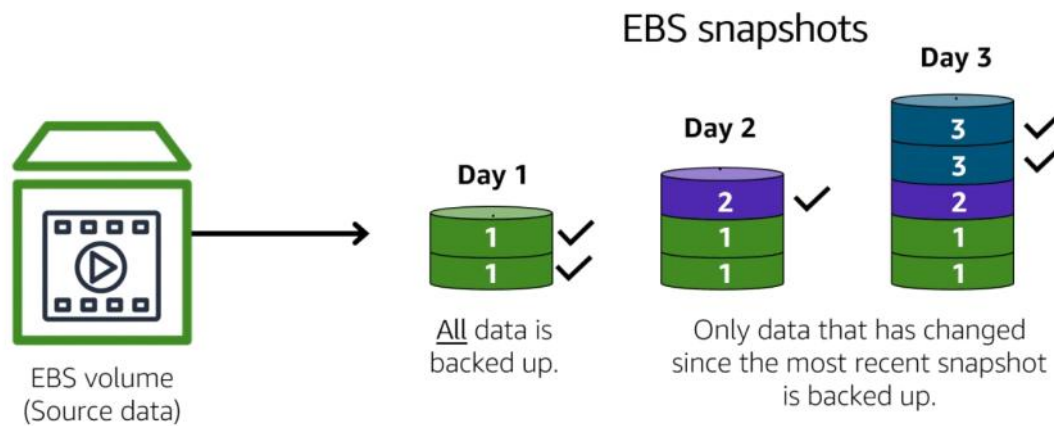
What are EBS Snapshots?

EBS snapshots are incremental backups. The first backup copies all data; subsequent backups only copy changed blocks, saving storage costs.

EBS snapshots

Day 3





What is Amazon S3?

Amazon S3 (Simple Storage Service) is a fully managed object storage service with **99.999999999% durability**. It can store any type of file up to **5 TB per object**, with virtually unlimited total capacity.

Buckets and Objects

A **bucket** is a container for objects with a globally unique name. An **object** consists of data (the file), metadata (information about it), and a key (unique identifier).

Everything in S3 is private by default. Grant access using:

- **Bucket policies:** Allow or deny actions on the bucket
- **Identity-based policies:** Control user, group, or role permissions
- **Encryption:** Protect data at rest and in transit

Lifecycle Management :

Automate moving objects between storage classes with lifecycle rules:

- **Transition actions:** Move objects to cheaper storage classes over time
- **Expiration actions:** Automatically delete objects after a set period

Amazon S3 Storage Classes :

S3 offers different storage classes based on how often you access your data and how fast you need it.

Frequently Accessed Data

Storage Class	Best For
S3 Standard	Data accessed often. Stored in 3+ AZs. Highest availability, highest cost.
S3 Express One Zone	Fastest access (single-digit ms). 10x faster than Standard. 1 AZ.

Infrequently Accessed Data

Storage Class	Best For
S3 Standard-IA	Data accessed less often. 3+ AZs. Lower storage cost, higher retrieval cost.
S3 One Zone-IA	Same as Standard-IA but stored in 1 AZ. Cheaper.
S3 Intelligent-Tiering	Unknown access patterns. Automatically moves data between tiers.

Archive Storage (Glacier)

Storage Class	Best For
Glacier Instant Retrieval	Rarely accessed, but need instant access. 68% cheaper than Standard-IA.

Glacier Flexible Retrieval	Accessed 1-2x per year. Retrieval: minutes to hours.
Glacier Deep Archive	Long-term storage (7-10+ years). Lowest cost. 12-hour retrieval.

What is Amazon EFS?

Amazon EFS (Elastic File System) is a fully managed, scalable file storage service using the NFS protocol. It automatically scales to petabytes and allows multiple EC2 instances to access the same file system simultaneously.

Amazon EFS Benefits

- **Multi-AZ redundancy:** Replicates data across Availability Zones for high availability
- **Shared access:** Thousands of concurrent NFS connections from multiple EC2 instances
- **Elastic storage:** Grows and shrinks automatically - pay only for what you use

When to Use Each

- **EBS:** Databases, boot volumes, applications needing fast read/write on a single instance
- **S3:** Static files, backups, media storage, data lakes, web content
- **EFS:** Shared file systems, content management, web serving across multiple instances

What is the lowest-cost, durable storage option for retaining database backups for immediate retrieval?

- A. Amazon S3**
- B. Amazon Glacier
- C. Amazon EBS
- D. Amazon EC2 Instance Store

Which of the following is a fast and reliable NoSQL database service?

- A. Amazon Redshift
- B. Amazon RDS
- C. Amazon DynamoDB**
- D. Amazon S3

Which storage service can be used as a low-cost option for hosting static websites?

- A. Amazon Glacier
- B. Amazon DynamoDB
- C. Amazon Elastic File System (Amazon EFS)
- D. Amazon Simple Storage Service (Amazon S3)**

Which service provides a virtually unlimited amount of online highly durable object storage?

- A. Amazon Redshift
- B. Amazon Elastic File System (Amazon EFS)**
- C. Amazon Elastic Container Service (Amazon ECS)**
- D. Amazon S3

Which AWS service should be used for long-term, low-cost storage of data backups?

- A. Amazon RDS
- B. Amazon Glacier**
- C. AWS Snowball
- D. AWS EBS

Which of the following options are NOT the throughput models available with Amazon EFS?

- ☒ Bursting Throughput This mode scales with the amount of storage in your file system.
- ☒ Provisioned Throughput Allows you to specify a consistent level of throughput independent of the file system's size
- ☒ Max I/O Throughput Not a recognized throughput model for Amazon EFS.
- ☒ Elastic Throughput Automatically scales the throughput based on the file system's activity.

A media company that uses AWS needs to transfer 2 TB of data. The organization is trying to estimate the data transfer cost.



Which type of transfer of the data would result in no cost for the company?

- ☒ Inbound data transfer from the internet **AWS does not charge for inbound data transfer**
- ☒ Outbound data transfer to the internet Charged per GB (e.g., \$0.09/GB for the first 10TB from S3).
- ☒ Data transfer between AWS Regions Charged both ways (e.g., \$0.02/GB between US-East and EU-West).
- ☐ Data transfer between Availability Zones Charged at \$0.01/GB (except for some services like S3 replication).

Transfer Type	Description	Cost (Approx.)
Inbound Data Transfer	Data transferred into AWS from the internet	Free across all services and regions
Outbound Data Transfer to Internet	Data transferred from AWS to the internet	Varies by region, typically starts at \$0.09/GB for the first 10 TB/month
Inter-Region Data Transfer	Data transferred between AWS regions (e.g., us-east-1 to eu-west-1)	\$0.02 to \$0.17/GB, depending on regions involved
Intra-Region (Same AZ)	Data transferred within the same Availability Zone	Free
Intra-Region (Cross-AZ)	Data transferred between Availability Zones in the same region	\$0.01/GB (typical for EC2 to EC2 or EC2 to RDS)
Via NAT Gateway	Data routed through a NAT Gateway within the same region	\$0.045/GB (data processing fee)

A media company needs fully managed, highly reliable, and scalable file storage that is accessible over the Server Message Block (SMB) protocol.

Which AWS service will meet these requirements?

- ☒ Amazon S3 Object storage (not file storage; no SMB support).
- ☒ Amazon Elastic File System (Amazon EFS) Linux-optimized file storage (uses NFS, not SMB)
- ☒ Amazon FSx for Windows File Server Amazon FSx for Windows File Server is a fully managed, highly reliable, and scalable file storage service that natively supports SMB protocol
- ☒ Amazon Elastic Block Store (Amazon EBS) Block storage for EC2 (not shared file storage).

A media company is storing media data that will not be frequently accessed in the AWS Cloud. If the company needs to access the data, the data needs to be retrieved within 12 hours. The company wants a solution that is cost-effective for storage costs for each gigabyte.

Which Amazon S3 storage class will meet these requirements?

- ☒ S3 Standard **S3 Standard** → High cost (**~\$0.023/GB/month**), not optimal to store infrequently accessed data
- ☒ S3 Glacier Flexible Retrieval Cost-effective storage (~\$0.0036/GB/month, cheaper than S3 Standard-IA or One Zone-IA). Retrieval times of **1 minute to 12 hours**.
- ☒ S3 One Zone-Infrequent Access (s3 One Zone-IA) Lower redundancy (single AZ) and higher cost (**~\$0.010/GB/month**) than Glacier.
- ☐ S3 Standard-Infrequent Access (s3 Standard-IA) Faster access (milliseconds) but higher cost (**~\$0.0125/GB/month**).

AWS Database Services

AWS offers database services for different needs: relational databases for structured data, NoSQL for flexible schemas, and specialized services for analytics and migration.

Database Types

Relational (SQL)

Relational databases have data stored in tables with rows and columns, linked by relationships.

Examples of relational database services:

- **Amazon RDS:** Managed MySQL, PostgreSQL, MariaDB, Oracle, SQL Server
- **Amazon Aurora:** High-performance MySQL/PostgreSQL-compatible

NoSQL

NoSQL databases have data stored in flexible schemas, horizontal scaling for high throughput.

Examples of NoSQL database services:

- **Amazon DynamoDB:** Key-value and document database

Analytics & Migration

Examples of analytics and migration services:

- **Amazon Redshift:** Data warehouse for analytics
- **AWS DMS:** Database migration with minimal downtime

What is Amazon RDS?

Amazon RDS (Relational Database Service) is a managed service for relational databases that use SQL. RDS handles backups, patching, and hardware provisioning automatically.

Supported database engines: Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, SQL Server

Common use cases are: web applications, enterprise workloads, e-commerce platforms

RDS Benefits

- **Multi-AZ deployment:** Replicates data to a standby instance for high availability
- **Automated backups:** Plus manual snapshots when needed
- **Read replicas:** Offload traffic for better performance
- **Security:** VPC isolation, encryption at rest and in transit

Amazon Aurora

Aurora is a **high-performance database** compatible with MySQL and PostgreSQL, delivering up to **5x MySQL throughput** and **3x PostgreSQL throughput**.

Throughput = how much data can be processed or transferred in a given amount of time.

Think of it as "how much work/data can be handled per second."

It creates six copies of data across three AZs, provides 99.99% availability, and automatically scales storage from 10 GB to 128 TB.

Aurora Benefits

- **Distributed storage:** Across multiple nodes for high performance
- **Continuous backup:** To S3 with point-in-time recovery
- **Auto-failover:** Redirects traffic to healthy replicas

Use cases: Gaming, media/content management, real-time analytics

What is Amazon DynamoDB?

DynamoDB is a fully managed NoSQL database for key-value and document data. NoSQL databases use flexible schemas where each item has a key and attributes, but unlike relational databases, different items can have different attributes. DynamoDB delivers **single-digit millisecond response times** at any scale by automatically spreading data across servers. Common use cases are: gaming platforms, financial services, mobile apps with global users

DynamoDB Benefits

- **Auto-scaling:** Throughput scales up or down based on usage
- **High availability:** 99.999% availability across three facilities per Region
- **Encryption:** All data automatically encrypted at rest

Comparing Amazon RDS and DynamoDB

RDS vs DynamoDB

RDS and DynamoDB serve different needs. RDS is for relational data with complex queries. DynamoDB is for simple key-value lookups at massive scale.

Comparison

Feature	Amazon RDS	Amazon DynamoDB
Type	Relational (SQL)	Non-relational (NoSQL)
Queries	Complex queries and joins	Simple key-value lookups
Schema	Fixed schema	Flexible schema
Latency	Milliseconds	Single-digit milliseconds

When to Use Each

- **RDS:** Complex queries, joins, ACID (Atomicity, Consistency, Isolation, Durability) transactions, traditional applications
- **DynamoDB:** High-traffic apps, gaming, IoT, real-time data at any scale

What is Amazon Redshift?

Amazon Redshift is a fully managed data warehouse service for big data analytics. It can analyze petabytes of data from multiple sources using SQL queries, ideal when data becomes too large for traditional databases.

Common use cases are: business intelligence, reporting dashboards, log analysis, and data lake analytics

What is AWS DMS?

AWS Database Migration Service moves data from a source database to a target database. Your source database remains operational during migration, minimizing downtime.

DMS supports a wide range of databases and is simple to use.

Use Cases

- Test applications against production data without affecting it
- Combine multiple databases into one
- Replicate data to other sources for analytics or backup

AWS Additional Database Services

1. Amazon DocumentDB

A fully managed, MongoDB-compatible document database that stores JSON-like documents with dynamic schemas. Ideal for content management, user profiling, and catalog management.

2. Amazon Neptune

A purpose-built graph database for storing and querying highly connected data. Supports property graph and RDF models. Use cases: social networks, fraud detection, and recommendation systems.

3. **Amazon Quantum Ledger Database (QLDB)**

A ledger database with an immutable, cryptographically verifiable transaction log. Tracks all data changes over time. Ideal for financial records, supply chain systems, and audit trails.

4. **AWS Managed Blockchain**

Create and manage blockchain networks using open-source frameworks like Ethereum and Hyperledger Fabric.

5. **Amazon ElastiCache**

A fully managed in-memory caching service that provides sub-millisecond latency. Supports Redis, Valkey, and Memcached. Use cases: session management, query caching, and gaming leaderboards.

6. **Amazon DynamoDB Accelerator (DAX)**

An in-memory cache specifically for DynamoDB that improves response times from milliseconds to microseconds. Fully managed with no code changes required.

Which AWS services should be used for read/write of constantly changing data? (Choose two.)

- A. Amazon Glacier
- B. Amazon RDS**
- C. AWS Snowball
- D. Amazon Redshift
- E. Amazon EFS**

What is one of the advantages of the Amazon Relational Database Service (Amazon RDS)?

- A. It simplifies relational database administration tasks.**
- B. It provides 99.9999999999% reliability and durability.
- C. It automatically scales databases for loads.
- D. It enabled users to dynamically adjust CPU and RAM resources.

A customer needs to run a MySQL database that easily scales.
Which AWS service should they use?

- A. Amazon Aurora**
- B. Amazon Redshift
- C. Amazon DynamoDB
- D. Amazon ElastiCache

Which AWS managed service is used to host databases?

- A. AWS Batch
- B. AWS Artifact
- C. AWS Data Pipeline
- D. Amazon RDS**

Amazon Relational Database Service (Amazon RDS) offers which of the following benefits over traditional database management?

- A. AWS manages the data stored in Amazon RDS tables.
- B. AWS manages the maintenance of the operating system.**
- C. AWS automatically scales up instance types on demand.
- D. AWS manages the database type.

Which service is best for storing common database query results, which helps to alleviate database access load?

- A. Amazon Machine Learning
- B. Amazon SQS
- C. Amazon ElastiCache**
- D. Amazon EC2 Instance Store

A company wants a time-series database service that makes it easier to store and analyze trillions of events each day.

Which AWS service will meet this requirement?

- ☒ Amazon Neptune **A graph database** service. Optimized for **relationships and connected data**, not time-series data.
- ☒ Amazon Forecast **A machine learning service for predictive forecasting**. Not a database; used for **forecasting future trends**, not storing time-series data.
- ☒ Amazon DynamoDB **A NoSQL key-value and document database**. Can store time-series data, but **not optimized** for time-series analytics at scale.
- ☐ Amazon Timestream **Purpose-built time-series database** service. Designed to **store and analyze trillions of time-stamped events** per day. Automatically handles **data lifecycle management** (e.g., recent vs. historical data).

What is AI and Machine Learning?

Artificial Intelligence (AI) enables computers to perform tasks that typically require human intelligence, while Machine Learning (ML) trains systems to learn from data and improve over time.

From e-commerce to medical research, organizations use AI and ML to analyze data and innovate. AWS AI/ML solutions help you predict trends, automate processes, and derive actionable insights.

Artificial Intelligence (AI)

A broad field focused on developing intelligent systems capable of performing humanlike tasks such as reasoning, learning, and problem-solving.

Machine Learning (ML)

A subset of AI that trains machines to perform complex tasks without explicit instructions. ML finds patterns in historical data to produce a model that makes predictions on new data.

Three Ways to Use AWS AI/ML Services

AWS offers three tiers, from simple to advanced:

- **AI Services:** Ready-to-use models that work immediately. No ML knowledge needed.
- **ML Services:** Build your own models with Amazon SageMaker AI. AWS manages the infrastructure.
- **ML Frameworks:** Full control to build custom models from scratch. Requires ML expertise.

AI Services

Pre-built models that work immediately. No ML knowledge required.

Service	What It Does
Amazon Comprehend	Extract insights from text (sentiment, key phrases, language)
Amazon Polly	Convert text to lifelike speech in multiple languages
Amazon Transcribe	Convert speech to text with speaker identification
Amazon Translate	Translate text between languages in real-time
Amazon Kendra	Intelligent search that understands natural language
Amazon Rekognition	Analyze images and videos (objects, faces, text, scenes)
Amazon Textract	Extract text from documents, forms, and tables
Amazon Lex	Build chatbots with voice and text interfaces
Amazon Personalize	Create personalized recommendations from historical data

Data Pipelines on AWS

AWS provides services for each stage of data pipeline: ingestion, storage, cataloging, processing, and analysis. These services integrate together to build complete data workflows.

Data Ingestion

Collect data and move it to storage. Use real-time when you need data immediately, batch when a small delay is acceptable.

Service	What It Does
Amazon Kinesis Data Streams	Collects large amounts of data in real-time from apps, devices, and sensors. Serverless and scales automatically.
Amazon Data Firehose	Collects data in near real-time (seconds). Fully managed - delivers data to storage automatically.

Data Storage

Centralize data from multiple sources for analysis.

Service	What It Does
Amazon S3	Object storage for data lakes. Stores any amount of data (structured or unstructured). Scales automatically.
Amazon Redshift	Data warehouse for structured data. Stores petabytes of data with pay-as-you-go pricing.

Data Cataloging

Organize and find your data across services.

Service	What It Does
AWS Glue Data Catalog	Central place to store information about your data. Makes it easy to find and use data across services.

Data Processing

Clean and transform data for analysis.

Service	What It Does
AWS Glue	ETL service (Extract, Transform, Load). Prepares data for analysis. Works with the Glue Data Catalog.
Amazon EMR	Large-scale data processing. Supports Spark, Hadoop, and Hive. AWS manages the infrastructure.

Analysis and Visualization

Query data and create visual insights.

Service	What It Does
Amazon Athena	Run SQL queries on data in S3. Serverless - pay only for queries you run.
Amazon Redshift	Fast SQL queries on large datasets. Best for frequent, complex analytics.
Amazon QuickSight	Create dashboards and reports. Amazon Q lets you ask questions in plain English.
Amazon OpenSearch Service	Search your data with keywords or natural language. Real-time dashboards.

Security is just the first step. You also need to monitor resources, verify everything works correctly, and follow regulations. AWS provides tools to detect problems early and respond quickly.

Cloud Management Fundamentals

- **Secure:** Protect data and systems from unauthorized access
- **Monitor:** Watch systems continuously to detect problems or threats
- **Audit:** Review security controls regularly to verify effectiveness
- **Compliance:** Follow industry rules and regulations

What is Monitoring?

Monitoring is the continuous process of collecting and tracking the health and performance of your AWS infrastructure, services, and applications. The goal is to ensure optimal performance and identify issues before they impact users.

Why Monitoring Matters

- **Security:** Detect threats or anomalies in real time
- **Reliability:** Track system health and availability
- **Performance:** Optimize resources based on usage patterns
- **Costs:** Monitor resource utilization and spending
- **Proactive response:** Fix issues before they impact users

Monitoring uses real-time tools, log analysis, and dashboards to give you visibility into your systems.

Amazon CloudWatch

CloudWatch monitors your AWS resources and applications in **real time**, providing visibility into resource utilization, application performance, and operational health.

CloudWatch Features

Metrics

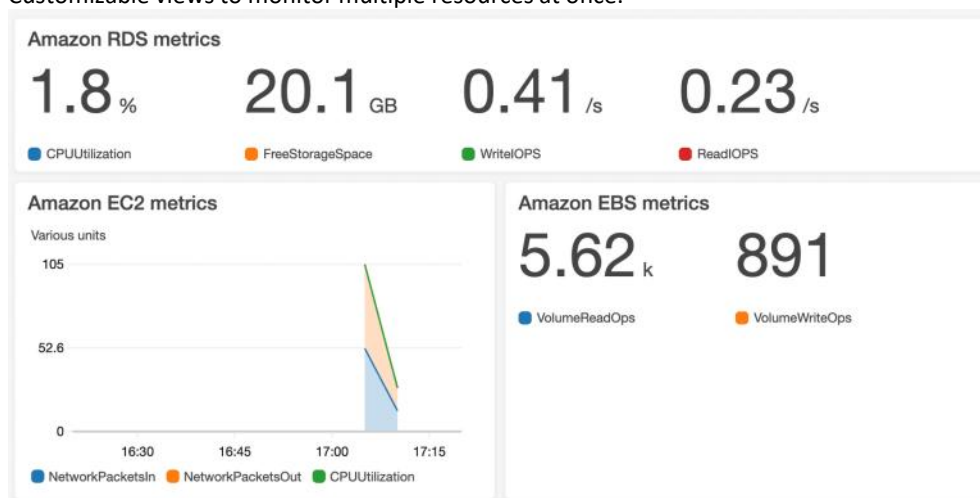
Collects metrics from AWS resources, applications, and on-premises servers.

Alarms

Define thresholds to trigger notifications or automatic actions. Example: stop idle services when utilization drops below a threshold.

Dashboards

Customizable views to monitor multiple resources at once.



AWS CloudTrail

CloudTrail tracks user activity and API usage across AWS, on-premises, and other cloud providers. It provides a detailed history of API calls so you can track changes, identify who made them, and when.

Use CloudTrail for auditing, security monitoring, troubleshooting, and proving compliance.

CloudTrail Features

Events

Captures details about actions in your AWS account, such as API calls, console actions, and other activities. Event history provides a viewable, searchable, and **immutable record of the past 90 days at no charge.**

Logs

Delivers event log files to your S3 bucket for long-term storage. Securely stored logs can prove compliance with regulations like PCI and HIPAA.

Insights

Analyzes your normal patterns of API call volume and error rates. Generates alerts when activity deviates from normal patterns, helping detect anomalous behavior and unusual activity.

What is Compliance?

Compliance ensures your cloud resources and data adhere to regulations, industry standards, and internal policies. AWS helps by inheriting security controls, providing third-party validation, and offering on-demand compliance reports.

AWS Artifact is a service from AWS that provides on-demand access to **AWS security and compliance documentation**, as well as tools for managing compliance agreements.

It is primarily used by organizations that need evidence of AWS's compliance with industry standards or that must review and accept legal agreements.

Key features include:

- **Compliance reports:** **Download audit reports and certifications, such as:**
 - SOC 1, SOC 2, and SOC 3 reports
 - ISO 27001, ISO 27017, and ISO 27018 certifications
 - PCI DSS compliance documentation
 - HIPAA and other regulatory documents (where applicable)
- **Agreements:** **Review, accept, and manage AWS legal agreements, such as:**
 - Business Associate Addendum (BAA) for HIPAA
 - Data Processing Addendum (DPA)
 - Other region- or service-specific agreements

AWS Security Hub

AWS Security Hub is a cloud security posture management service that provides a comprehensive view of your security state in AWS. It helps you assess your AWS environment against security industry standards and best practices.

Why use AWS Artifact?

- To provide compliance evidence during audits.
- To verify that AWS meets your organization's regulatory requirements.
- To simplify governance and risk management by accessing compliance documents in one place.

What is Auditing?

Auditing means checking and verifying whether something is being done correctly according to rules, policies, or standards.

Auditing Services

1. AWS Config

Assess, audit, and evaluate the configurations of your AWS resources. Config tracks configuration changes and evaluates them against your desired state.

2. AWS Audit Manager

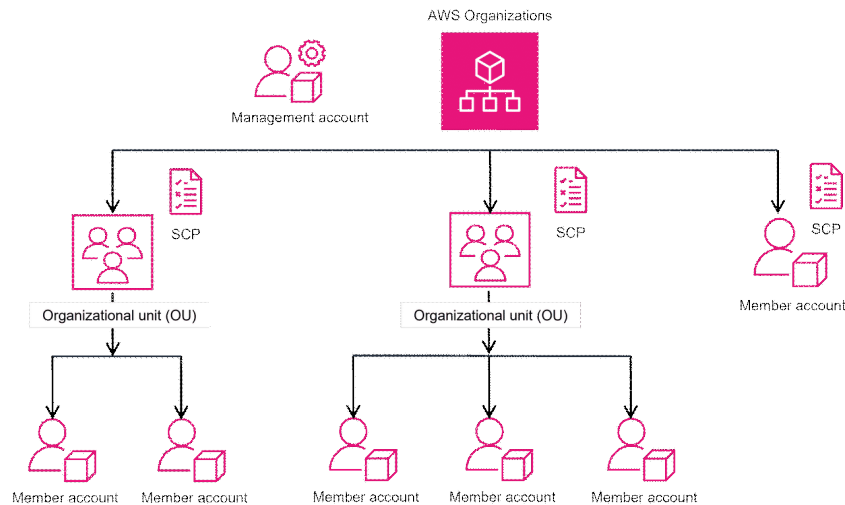
Continually audits your AWS usage to simplify risk and compliance assessment. Helps collect evidence and manage audit data automatically.

AWS Organizations

As companies grow, managing multiple AWS accounts becomes challenging. AWS Organizations helps you centrally manage and govern your environment, automate account creation, and apply policies across groups of accounts.

Benefits and Use Cases

- Quickly scale by programmatically creating new AWS accounts
- Simplify permission management through Service Control Policies (SCPs)
- Manage and optimize costs across accounts
- Provide tools and access for security teams
- Share common resources across accounts



Management Account

The central AWS account that creates and manages the organization. When you create an organization, it automatically creates a root as the parent container for all accounts. The management account is responsible for overall control and governance.

Organizational Unit (OU)

A logical grouping of accounts within an AWS Organization. OUs can contain member accounts or nested OUs, allowing you to create a hierarchy that matches your business structure.

Service Control Policies (SCP)

Policies that restrict which AWS services, resources, and API actions users and roles in each account can access. SCPs can be applied to OUs or individual member accounts.

Member Account

Accounts that belong to the organization. They can be placed under an OU or directly under the root. Even accounts not in an OU benefit from features like consolidated billing.

What is AWS Trusted Advisor?

Optimizing large-scale cloud deployments is an ongoing process. **Trusted Advisor continuously evaluates your AWS environment using best practice checks across five categories:** cost optimization, performance, security, fault tolerance, and service limits. All AWS Support plans include access to dozens of checks. Business Support and advanced plans provide hundreds of checks.

Trusted Advisor Features

Check Categories

- **Cost optimization:** Identify unused resources and opportunities to reduce spending
- **Performance:** Improve speed and responsiveness of your applications
- **Security:** Review permissions and identify security gaps
- **Fault tolerance:** Increase availability and redundancy
- **Service limits:** Track usage against service quotas

Dashboard

Access Trusted Advisor from the AWS Management Console. The dashboard shows completed checks and results per category:

Cost Optimization



0 9 0

\$7,516.85

Potential monthly savings

Performance



3 7 0

Security



2 4 11

Fault Tolerance



0 15 5

Service Limits



37 0 1

Which of the following are features of Amazon CloudWatch Logs? (Choose two.)

- A. Summaries by Amazon Simple Notification Service (Amazon SNS)
- B. Free Amazon Elasticsearch Service analytics
- C. Provided at no charge
- D. Real-time monitoring**
- E. Adjustable retention**

If a customer needs to audit the change management of AWS resources, which of the following AWS services should the customer use?

- A. AWS Config**
- B. AWS Trusted Advisor
- C. Amazon CloudWatch
- D. Amazon Inspector

Which service allows a company with multiple AWS accounts to combine its usage to obtain volume discounts?

- A. AWS Server Migration Service
- B. AWS Organizations**
- C. AWS Budgets
- D. AWS Trusted Advisor

An organization is using AWS CloudTrail to monitor user activities and API calls in their AWS cloud workloads. Due to a regulatory requirements the organization must retain the CloudTrail logs for a longer duration, beyond CloudTrail's default retention settings. How can this be achieved?

- ☒ Create a Service request and request AWS support team to modify the default retention timelines **Does not support modifying default retention timelines through service requests.**
- ☒ Create an encrypted Amazon S3 and then enable CloudTrail event logging for the S3 buckets from S3 console. **Creating an encrypted S3 bucket is correct, enabling CloudTrail event logging must be done from the CloudTrail console.**
- ☒ Create an encrypted Amazon S3 and then enable CloudTrail event logging for the S3 buckets from CloudTrail console **By configuring CloudTrail to log events directly to the S3 bucket, you can retain the logs for as long as needed by setting appropriate lifecycle policies on the bucket.**
- ☐ Use Amazon CloudWatch to store the user activities and API call logs **CloudWatch is not designed for long-term storage of CloudTrail logs.**

Your organization recently faced some destructive event of deleting production resources. You were tasked to determine which user has deleted the resources, and from which IP address the request was initiated.

Which AWS service should you use for this activity?

- ☒ Amazon CloudWatch Used for monitoring and logging metrics related to AWS resources. Does not track detailed user activities or API calls
- ☒ Amazon GuardDuty Threat detection service that continuously monitors for malicious activity and unauthorized behavior. Does not provide detailed logs of user activities or API calls.
- ☒ Amazon S3 Storage Analytics S3 Storage Analytics provides insights into storage usage and access patterns for S3 buckets. It does not track user activities or API calls related to resource deletions.
- ☐ AWS CloudTrail Records AWS API calls and user activities within your AWS account. It provides detailed logs that include information about who made the API call, what actions were taken, and from which IP address the request originated.



Which of the following are benefits of using AWS Trusted Advisor? (Choose two.)

- ☒ Providing high-performance container orchestration Container orchestration is handled by Amazon EKS/ECS, not Trusted Advisor.
- ☒ Creating and rotating encryption keys Encryption keys are managed by AWS KMS, not Trusted Advisor.
- ☒ Detecting underutilized resources to save costs **Cost Optimization** (e.g., identifying idle EC2 instances, underutilized EBS volumes) .
- ☒ Improving security by proactively monitoring the AWS environment Offers security checks that help you identify potential security gaps and vulnerabilities.
- ☐ Implementing enforced tagging across AWS resources Enforced tagging requires **AWS Organizations + Tag Policies**, not Trusted Advisor.



The AWS Free Tier lets you try services for free for the specified period.

It has three different offerings:

- Always Free
- 12 Months Free
- Trial

AWS Pricing Overview

1. Pay As You Go

With pay as you go, you can adapt to changing business needs and reduce the risk of overprovisioning or missing capacity. Pay only for the resources that you use.

2. Save When You Commit

For certain services, such as Compute services on AWS, Savings Plans offer savings over On-Demand prices when you commit to a 1-year or 3-year plan.

Driving Factors of Cost

There are three fundamental drivers of cost with AWS:

1. Compute

For compute resources, you pay by a certain span of time, like by the hour or by the second.

Unless you've made a reservation for which the cost is agreed upon beforehand, you pay from the time you launch a resource until the time you stop the instance.

2. Storage

Pricing for storage largely depends on how much storage you have provisioned or how much you are using.

For some storage options, such as Amazon S3, storage cost is tiered. This means you can optimize storage costs based on how frequently and quickly you need to access data.

3. Data Transfer

In most cases, there is no charge for inbound data transfer or for data transfer between AWS services within the same Region.

Outbound data transfer is aggregated across services and then charged at the outbound data transfer rate.

The more data you transfer, the less you pay per gigabyte.

AWS Pricing and Billing Services Overview

AWS provides services to help you manage and optimize your cloud costs. This includes consolidated billing through AWS Organizations, as well as tools to view, track, and forecast expenses.

AWS Organizations

AWS Organizations provides centralized management and governance of your AWS environment.

Using AWS Organizations, you can create, group, and manage accounts. You can also apply security policies at the account level and consolidate billing with multiple accounts using a single payment method.

Use Cases

- Consolidate multiple AWS accounts into one central organization
- Implement organization-wide policies

AWS Billing and Cost Management Dashboard

The AWS Billing and Cost Management dashboard centralizes cost management, showing current charges, usage, forecasts, and

detailed breakdowns.

It also provides tools to manage payments, view invoices, set budgets, and consolidate billing.

Use Cases

- Use helpful visualizations and billing reports of monthly AWS spend
- Set up and manage payment methods

AWS Budgets

AWS Budgets helps set custom budgets and sends alerts when costs, usage, or Savings Plans and Reserved Instances (RIs) utilization or coverage exceed defined thresholds.

AWS Cost Explorer

AWS Cost Explorer helps visualize, analyze, and manage AWS costs and usage with interactive graphs, reports, and forecasts. It provides insights into spending patterns, trends, and Reserved Instance recommendations.

AWS Pricing Calculator

The AWS Pricing Calculator is a web-based planning tool that you can use to create estimates.

The screenshot shows the AWS Pricing Calculator interface. At the top, there's a dark blue header with the AWS logo, 'pricing calculator', and links for 'Feedback', 'English', and 'Contact Sales'. Below the header, the breadcrumb trail reads 'AWS Pricing Calculator > My Estimate > Add Amazon EC2'. The main content area is titled 'Configure Amazon EC2' with an 'Info' link. On the left, a sidebar shows 'Step 1: Select service' and 'Step 2: Configure Amazon EC2'. The 'Region' dropdown is set to 'US East (Ohio)'. There are two radio button options: 'Quick estimate' (selected) and 'Advanced estimate'. The 'Quick estimate' description says: 'Choose this option for fast and easy route to a ballpark estimate based on minimum requirements or a specific instance search. The estimate assumes consistent utilization.' The 'Advanced estimate' description says: 'Choose this option for a more detailed estimate that accounts for workload, data transfer costs, additional storage options, and other, less common instance requirements. For example, you know that you get a lot of traffic on Mondays but not much traffic throughout the rest of the week, and you want an estimate that takes this workload into account.' Below these options is a section for 'EC2 instance specifications' with an 'Info' link. Under 'Operating system', there's a dropdown menu currently showing 'Linux'.

Use Cases

- Estimate potential costs before deployment
- Compare costs of different AWS services and configurations

AWS Support Plans

Types of AWS Support

1. Basic Support

Included for all AWS customers

- Access to documentation, whitepapers, and AWS re:Post
- Core AWS Trusted Advisor checks
- Technical Account Management not included

2. Developer Support

Recommended for experimenting or testing in AWS

- Response times: < 24 hours for general guidance, < 12 hours when systems impaired
- Best practice guidance
- Client-side diagnostic tools
- Building-block architecture support

3. Business Support

Recommended minimum tier for production workloads in AWS

- Response times include: < 4 hours when production system impaired, < 1 hour when production system is down
- Full set of AWS Trusted Advisor checks
- Use-case guidance
- Limited support for third-party software

4. Enterprise On-Ramp Support

Recommended for production and business critical workloads in AWS

- Response times: < 30 minutes when business-critical system is down
- Full set of AWS Trusted Advisor checks
- A pool of technical account managers (TAMs) provide proactive guidance

5. Enterprise Support

Recommended for business critical and mission critical workloads in AWS

- Response times: < 15 minutes when business- or mission-critical system is down
- Full set of AWS Trusted Advisor checks and prioritized recommendations
- A designated TAM provides consultative architectural and operational guidance

Technical Account Manager (TAM)

A TAM is included with the Enterprise On-Ramp and Enterprise Support plans.

The TAM serves as your primary AWS contact, offering expert guidance on:

- Using AWS services
- Optimizing architectures
- Managing costs
- Connecting you with AWS programs and experts

AWS Marketplace

The AWS Marketplace is a digital catalog that includes thousands of software listings from independent software vendors. You can use AWS Marketplace to find, test, and buy software that runs on AWS.

For each listing, you can access detailed information on pricing options and reviews from other AWS customers.

AWS Partner Network

The AWS Partner Network (APN) is a global community that uses AWS technologies, programs, expertise, and tools to build solutions and services for customers.

Which service should a customer use to consolidate and centrally manage multiple AWS accounts?

- A. AWS IAM
- B. AWS Organizations**
- C. AWS Schema Conversion Tool
- D. AWS Config

A customer is using multiple AWS accounts with separate billing.

How can the customer take advantage of volume discounts with minimal impact to the AWS resources?

- A. Create one global AWS account and move all AWS resources to the account.
- B. Sign up for three years of Reserved Instance pricing up front.
- C. Use the consolidated billing feature from AWS Organizations.**
- D. Sign up for the AWS Enterprise support plan to get volume discounts.

Where should a company go to search software listings from independent software vendors to find, test, buy and deploy software that runs on AWS?

- A. **AWS Marketplace**
- B. Amazon Lumberyard
- C. AWS Artifact
- D. Amazon CloudSearch

Which AWS support plan includes a dedicated Technical Account Manager?

- A. Developer
- B. **Enterprise**
- C. Business
- D. Basic

If each department within a company has its own AWS account, what is one way to enable consolidated billing?

- A. Use AWS Budgets on each account to pay only to budget.
- B. Contact AWS Support for a monthly bill.
- C. **Create an AWS Organization from the payer account and invite the other accounts to join.**
- D. Put all invoices into one Amazon Simple Storage Service (Amazon S3) bucket, load data into Amazon Redshift, and then run a billing report.

An organization wants register for AWS Support plan. However, the organization reached to you to clarify different AWS support plans.

At what support level do users receive access to a support concierge?

A dedicated point of contact for any account or billing-related issues.

- ☒ Basic Support Free, but only includes account/billing help (no concierge)
- ☒ Developer Support Business-hour email access (no concierge/TAM).
- ☒ Business Support 24/7 tech support but lacks a dedicated concierge.
- ☐ Enterprise Support Enterprise Support is the only AWS Support plan that includes a Support Concierge, which provides, Dedicated technical account manager (TAM) for proactive guidance. Priority 24/7 access to senior cloud support engineers.

An online retail company has purchased AWS Enterprise support plan.

How does the AWS Enterprise Support Concierge team help users?

- ☒ Supporting application development
Handled by Professional Services or Technical Account Managers (TAMs).
- ☒ Providing architecture guidance
Provided by TAMs/Solutions Architects.
- ☒ Answering billing and account inquiries
AWS Enterprise Support Concierge team specializes in Billing and account assistance and Operational tasks
- ☒ Answering questions regarding technical support cases
Managed by AWS Support Engineers (not Concierge).

Cost Optimization

Cost optimization involves continuously refining your AWS environment to reduce spending while maintaining performance. This includes rightsizing resources, using appropriate pricing models, and leveraging AWS tools to identify savings opportunities.

Cost Optimization Strategies

Here are practical ways to optimize costs across common AWS services.

1. Amazon EC2 Optimization

One way to optimize cost with Amazon EC2 instances is to **rightsized** your resources. This means that you analyze and adjust your resources to meet the needs of your workload. Services such as AWS Compute Optimizer can help rightsize your compute resources.

Spot Instances

Using Spot Instances can help optimize for cost.

Spot Instances work well for workloads that are tolerant to interruptions.

Spot Instances use spare Amazon EC2 capacity for a significant discount compared to On-Demand instances.

2. Auto Scaling

Another way to help optimize cost with compute resources is auto scaling.

When demand drops, AWS Auto Scaling will automatically remove any excess resource capacity, so you avoid overspending. Application load balancing also helps distribute traffic across EC2 instances.

3. Amazon RDS Optimization

Rightsizing is an important piece of cost optimization in your Amazon RDS instances.

Amazon RDS can scale storage using storage autoscaling which prevents over- or underprovisioning.

4. Read Replicas

Read replicas scale capacity horizontally, meaning your resources won't need to scale up into a larger instance.

Instead of scaling the primary instance, read replicas are used to store data and can be used for heavy read workloads which reduces the strain on the primary instance.

5. Caching

Using something like Amazon ElastiCache for caching can also reduce the load on your primary instance and optimize for cost.

6. Amazon S3 Optimization

Using the right storage class is key to optimizing cost in the cloud.

Storage Classes

- **S3 Glacier Deep Archive:** Low-cost storage tier ideal for data that is accessed once or twice a year
- **S3 Intelligent-Tiering:** Great choice for data with unknown or changing access patterns

7. VPC Endpoints

You can use VPC endpoints to help optimize for cost as well.

Using VPC endpoints for Amazon S3 access can help reduce data transfer costs.

Introduction to Migration

Cloud migration is the process of moving your organization's digital assets, IT resources, applications, and databases from on-premises infrastructure to the AWS Cloud.

Three Phases of the Migration Process

AWS helps guide companies through the migration process, which can be divided into three sequential phases. From the first phase to the last, there are AWS services and tools to support your migration.

1. Assess

In this phase, you build the business case for the migration and assess your readiness. One of the services used in this phase is the **Migration Evaluator**.

2. Mobilize

In this phase, you prepare the organization and mobilize the resources needed for the migration. Two services you might use in this phase are **AWS Application Discovery Service** and the **AWS Migration Hub**.

3. Migrate and Modernize

In this phase, you use your strategy, plan, and the best practices to migrate and modernize.

Tools to support you include:

- AWS Application Migration Service
- AWS Database Migration Service (AWS DMS)
- AWS DataSync
- AWS Transfer Family
- AWS Snow Family

While AWS Migration Hub helps you mobilize your resources, you can use it as a one-stop place to also manage during the migrate and modernize phase.

The Seven Rs

When a company migrates to the AWS Cloud, they have many options. There are seven migration strategies to help them reach their goals, also known as the **seven Rs**:

1. Relocate
2. Rehost
3. Replatform
4. Refactor
5. Repurchase
6. Retain
7. Retire

The company may choose certain strategies depending on its specific needs, application types, where the company starts, and the company's future plans.

Database Migrations

Migrating Databases

Migrating your database to the cloud provides an opportunity to redesign and improve your database architecture. You might also consider migrating to an AWS managed database service or an open source database to reduce licensing costs.

Same Engine vs Different Engine

Migrating to the **same database engine** (e.g., MySQL to Amazon RDS for MySQL) is straightforward.

Migrating to a **different engine** (e.g., Oracle to Amazon Aurora PostgreSQL) is more complex and may require application changes.

AWS calls these *homogeneous* (same engine) and *heterogeneous* (different engine) migrations.

AWS provides services to help with both types.

AWS Database Migration Service (AWS DMS)

AWS DMS makes it possible to quickly and securely migrate databases and perform ongoing data replication tasks for live databases and data warehouses.

AWS Schema Conversion Tool (AWS SCT)

AWS SCT converts the schema and database code (tables, data types, procedures, etc.) so it works with the new database engine.

Transferring Data Online

Considerations for Data Migration

When you migrate data to the AWS Cloud, there are a few considerations to keep in mind:

- **Security:** Will it get there safely?
- **Data validation:** Will it get there in one piece?
- **Scheduling:** When is the best time?
- **Bandwidth requirements:** Do you have enough?

For the majority of data migration workloads, AWS DataSync will do the job.

AWS DataSync

AWS DataSync is specifically designed for automating and accelerating data transfer.

DataSync simplifies and accelerates moving large amounts of data between on-premises storage and AWS storage services like Amazon S3.

It automates many aspects of the transfer process, including running instances, encryption, and network optimization.

AWS Direct Connect

AWS Direct Connect establishes a dedicated private connection between your network and your VPC in the AWS Cloud. Because it bypasses the public internet, it provides a fast, reliable, and secure way to transfer large amounts of data.

Transferring Data Offline

1. AWS Snowball Edge Storage Optimized Devices

AWS Snowball Edge Storage Optimized devices are a great solution for offline data migration where connecting to the internet might not be an option.

AWS Snow Family Devices

AWS Snow Family is a group of physical devices that transport data in and out of AWS, capable of transferring up to exabytes of data.

Device Types

Snowball Edge Storage Optimized	Snowball Edge Compute Optimized
Great for large-scale data migrations	Great for services that require a large amount of computing resources
80 TB of HDD storage space for object storage	42 TB of HDD storage for object storage

Transfer Type	Description	Cost (Approx.)
Inbound Data Transfer	Data transferred into AWS from the internet	Free across all services and regions
Outbound Data Transfer to Internet	Data transferred from AWS to the internet	Varies by region, typically starts at <u>\$0.09/GB</u> for the first 10 TB/month
Inter-Region Data Transfer	Data transferred between AWS regions (e.g., us-east-1 to eu-west-1)	\$0.02 to \$0.17/GB, depending on regions involved
Intra-Region (Same AZ)	Data transferred within the same Availability Zone	Free
Intra-Region (Cross-AZ)	Data transferred between Availability Zones in the same region	<u>\$0.01/GB</u> (typical for EC2 to EC2 or EC2 to RDS)
Via NAT Gateway	Data routed through a NAT Gateway within the same region	<u>\$0.045/GB</u> (data processing fee)

One of the advantages to moving infrastructure from an on-premises data center to the AWS Cloud is:

- A. it allows the business to eliminate IT bills.
- B. it allows the business to put a server in each customer's data center.
- C. it allows the business to focus on business activities.**
- D. it allows the business to leave servers unpatched.

Which AWS service allows companies to connect an Amazon VPC to an on-premises data center?

- A. AWS VPN
- B. Amazon Redshift
- C. API Gateway
- D. Amazon Direct Connect**

A company plans to use an Amazon Snowball Edge device to transfer files to the AWS Cloud.

Which activities related to a Snowball Edge device are available to the company at no cost?

- ☒ Use of the Snowball Edge appliance for a 10-day period
Initial 10-day period of using the Snowball Edge appliance is included in the service, it is not entirely free
- ☒ The transfer of data out of Amazon S3 and to the Snowball Edge appliance
AWS charges for data transfer out of Amazon S3.
- ☒ The transfer of data from the Snowball Edge appliance into Amazon S3
AWS does not charge for transferring data from the device into Amazon S3.
- ☐ Daily use of the Snowball Edge appliance after 10 days
After the initial 10-day period, there are additional charges for daily use of the Snowball Edge appliance.

A company plans to migrate its application from on premises to the AWS Cloud. The company needs to gather usage and configuration data for the application components.

Which AWS service will meet these requirements?

- ☒ AWS Database Migration Service (AWS DMS)
Migrates databases, doesn't gather system-level data.
- ☒ AWS Transfer Family
Securely transfers files (SFTP, FTP), unrelated to discovery.
- ☒ AWS Application Discovery Service
AWS Application Discovery Service automatically collects configuration, performance, and usage data from on-premises servers to help plan cloud migrations.
- ☐ AWS Global Accelerator
Improves network performance, not migration planning.

A company wants to migrate its server-based applications to the AWS Cloud. The company wants to determine the total cost of ownership for its compute resources that will be hosted on the AWS Cloud.

Which combination of AWS services or tools will meet these requirements? (Choose two.)

- ☒ AWS Pricing Calculator
Allows you to estimate costs for AWS services (e.g., EC2, RDS, S3) based on usage.
- ☒ Migration Evaluator.
Analyzes on-premises workloads and recommends cost-optimized AWS configurations.
- ☒ AWS Support Center. For technical support, not cost analysis.
- ☒ AWS Application Discovery Service. Discovers server dependencies, collects configuration, performance, and usage data from on-premises servers but doesn't calculate costs.
- ☐ AWS Database Migration Service (AWS DMS).
Migrates databases, unrelated to TCO.

AWS Well-Architected Framework

The Well-Architected Framework helps you design cloud solutions that are optimized, not just functional. It provides a structured approach to building secure, high-performing, resilient, and efficient architectures based on AWS best practices.

Design Principles of AWS Well-Architected Framework Reliability Pillar:

- Automatically recover from failure
- Test recovery procedures
- Scale horizontally to increase aggregate workload availability
- Stop guessing capacity
- Manage change through automation

The Six Pillars

The Well-Architected Framework is organized into **six key pillars**:

1. Operational Excellence
2. Security
3. Reliability
4. Performance Efficiency
5. Cost Optimization
6. Sustainability

Pillar	Easy meaning	Remember
1. Operational Excellence	Run and improve your system efficiently	Operate & improve
2. Security	Protect your data and resources	Protect
3. Reliability	Keep the system available and recover from failures	Keep running
4. Performance Efficiency	Use the right resources to get the required performance	Perform well
5. Cost Optimization	Avoid unnecessary spending and use resources efficiently	Save money
6. Sustainability	Reduce energy usage and environmental impact	Reduce waste

AWS Well-Architected Tool

The AWS Well-Architected Tool (AWS WA Tool) is a free service that helps assess and improve cloud workloads based on the six key pillars.

It offers:

- Workload reviews
- Milestone tracking
- Custom lenses for tailored evaluations
- Improvement plans

AWS Specialized Services

Beyond core infrastructure, AWS offers specialized services for specific use cases: development tools, business applications, end-user computing, and IoT.

Development Services

AWS offers several services to help developers automate CI/CD pipelines, monitor and debug applications, build GraphQL APIs, and deploy web and mobile applications on AWS.

1. AWS CodeBuild

CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces software packages for deployment.

It automatically scales to meet demand, and you only pay for the build time that you use.

2. **AWS CodePipeline**

CodePipeline is a fully managed CI/CD service that automates the build, test, and deploy phases of your release process. CI/CD pipelines automate the integration, testing, and deployment of code changes for quick and reliable software delivery.

3. **AWS X-Ray**

X-Ray is a powerful tracing, debugging, and performance analysis tool that helps developers visualize application behavior. With X-Ray, developers can quickly identify performance bottlenecks, troubleshoot issues, and optimize applications for better efficiency and reliability.

4. **AWS AppSync**

AWS AppSync is a fully managed GraphQL service. GraphQL is a query language for APIs where clients request only the specific data they need. With AppSync, developers can create a single API that securely accesses and combines data from multiple sources.

5. **AWS Amplify**

Amplify helps you streamline the process of developing, deploying, and managing secure and scalable full-stack applications on AWS.

With Amplify, developers can quickly add features like authentication, APIs, storage, and hosting with minimal infrastructure management.

Business Application Services

These services are ideal for managing business application needs such as customer service operations and email promotions.

1. **Amazon Connect**

Businesses can use this AI-powered contact center service to efficiently set up and operate a scalable customer service call center.

Amazon Connect provides capabilities for call routing, recording, and analytics while integrating seamlessly with other AWS services.

2. **Amazon Simple Email Service (Amazon SES)**

Amazon SES is a scalable and cost-effective email service provider that can be integrated into any application for reliable, high-volume email automation.

It helps businesses optimize the delivery of transactional and marketing emails, resulting in enhanced customer engagement.

End-User Computing Services

In modern businesses, IT departments often need to provide remote access to resources like virtual desktops and applications.

1. **Amazon AppStream 2.0**

AppStream 2.0 is a fully managed service that streams applications from the cloud directly to any compatible device.

AWS CAF - Cloud Adoption Framework

AWS CAF has six focus areas, also called **perspectives**.

Perspective	Easy meaning	Remember	Common roles
 Business	Why should the company move to AWS?	Money + Business goals	Business managers, Finance managers, Budget owners, Strategy stakeholders
 People	Do employees have the right skills?	People + Skills + Training	HR, Staffing, People managers
 Governance	What rules and processes should we follow?	Rules + Control	CIO, Program managers, Enterprise architects, Business analysts, Portfolio managers
 Platform	How will we build/migrate our IT systems?	Technology + Architecture	CTO, IT managers, Solutions architects
 Security	How will we protect our systems and data?	Protect + Audit + Control	CISO, IT security managers, IT security analysts
 Operations	How will we run the systems every day?	Run + Monitor + Recover	IT operations managers, IT support managers

What approach to transcoding a large number of individual video files adheres to AWS architecture principles?

- A. Using many instances in parallel
- B. Using a single large instance during off-peak hours
- C. Using dedicated hardware
- D. Using a large GPU instance type

When architecting cloud applications, which of the following are a key design principle?

- A. Use the largest instance possible
- B. Provision capacity for peak load
- C. Use the Scrum development process
- D. Implement elasticity

Which of the following is an important architectural design principle when designing cloud applications?

- A. Use multiple Availability Zones.
- B. Use tightly coupled components.
- C. Use open source software.
- D. Provision extra capacity.

A company is learning about the perspectives of the AWS Cloud Adoption Framework (AWS CAF).

Which perspective of the AWS CAF addresses the strategy management capability?

- ☐ Business Perspective
- ☐ People Perspective
- ☐ Governance Perspective
- ☐ Operations Perspective

Focuses on aligning cloud adoption with organizational goals, financial planning, and value realization.

Key Capabilities in the Business Perspective:

- ✓ Cloud strategy development
- ✓ Business case justification
- ✓ ROI measurement
- ✓ Stakeholder alignment

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A company wants to run its workloads in the AWS Cloud effectively, reduce management overhead, and improve processes.

Which AWS Well-Architected Framework pillar represents these requirements?

☒ Reliability

Fault tolerance and recovery (e.g., multi-AZ deployments).

☒ Operational excellence

Running workloads effectively | Reducing management overhead | Improving processes

☒ Performance efficiency

Optimizing compute/storage (e.g., right-sizing).

☐ Cost optimization

Protecting Networks and Applications

AWS provides built-in infrastructure protection and specialized security services to prevent and mitigate this unwanted traffic

Common Attack Types

DoS (Denial of Service)

An attacker floods a web application with excessive traffic, preventing legitimate requests from being served.

DDoS (Distributed Denial of Service)

Multiple infected computers (zombie bots) send excessive traffic simultaneously, making attacks harder to block.

Infrastructure Protection

AWS automatically protects against low-level attacks through its global infrastructure spanning Regions, Availability Zones, and edge locations.

- **Security Groups:** Filter traffic at the network level using Region-wide capacity
- **Elastic Load Balancing:** Handles traffic before your servers, preventing overload
- **AWS Regions:** Massive capacity makes them extremely difficult to overwhelm

Protection Services

AWS Shield

Standard: Free, automatic protection against common DDoS attacks with real-time detection and mitigation.

Advanced: Paid service with detailed attack diagnostics, sophisticated DDoS protection, and integration with CloudFront, Route 53, and ELB. Can combine with AWS WAF for custom rules.

AWS WAF

Web application firewall that monitors incoming requests and checks IP addresses against access control lists (ACLs). Blocks requests from known malicious sources while allowing legitimate traffic.

SQL Injection attack se bhi bachata hain

AWS Data Protection

Data Protection on AWS

Encryption protects data by making it unreadable without a key. AWS provides encryption for **data at rest** (stored in databases, S3) and **in transit** (moving between locations using SSL/TLS).

Data Protection Services

Service	What It Does
AWS KMS	Create and manage encryption keys. Control who can use them. Keys never leave AWS.
Amazon Macie	Uses ML to find sensitive data in S3. Helps meet compliance requirements.
AWS Certificate Manager	Manage SSL/TLS certificates for encrypting data in transit.

Detection and Response on AWS

Detection finds security threats. Response is how you investigate and fix them. AWS provides integrated services that find vulnerabilities, detect threats, investigate issues, and centralize security management.

Detection and Response Services

Service	What It Does
Amazon Inspector	Scans EC2, containers, and Lambda for vulnerabilities. Gives recommendations for fixes.
Amazon GuardDuty	Detects threats using ML. Monitors network activity and known malicious IPs.
Amazon Detective	Investigate threats after detection. Visualizations help find the root cause.
AWS Security Hub	Central dashboard for all security findings. Automates fixes.

AWS GuardDuty :

Intelligent threat detection service that continuously monitor AWS env for malicious activity and unauthorised behaviour.

IAM Access Analyzer ?

IAM Access Analyzer AWS ki ek security service hai jo help karti hai identify karne mein ki **AWS resources** kis ke saath access/share ho rahe hain, especially **outside your AWS account**.

An organization wants to store API keys, database credentials, securely in AWS. The organization is looking for a capability to automatically rotate the credentials without any manual intervention.

Which AWS service provides a centralized location for storing, managing, and retrieving sensitive information?

- ☒ AWS Systems Manager (SSM) - Parameter Store
SSM Parameter Store can store configuration & secrets, but doesn't offer automatic credentials rotation.
- ☒ Amazon DynamoDB
DynamoDB is a NoSQL database service used for storing and retrieving data. It is not designed for managing secrets or automatically rotating credentials.
- ☒ Amazon Macie
Security service that uses machine learning to discover, classify, and protect sensitive data in AWS.
- ☐ AWS Secrets Manager
Securely store, manage, and retrieve sensitive information such as database credentials, API keys, and other secret values. It provides the capability to automatically rotate credentials without manual intervention.

A company has deployed applications on Amazon EC2 instances. The company needs to assess application vulnerabilities and must identify infrastructure deployments that do not meet best practices.

Which AWS services can the company use to meet these requirements?

- ☒ AWS Trusted Advisor
Provides recommendations to optimize your AWS environment in terms of cost, performance, security, and fault tolerance.
- ☒ Amazon Inspector
Helps you identify vulnerabilities in your applications and infrastructure. It performs automated security assessments of your AWS resources
- ☒ AWS Config
Enables you to assess, audit, and evaluate the configurations of your AWS resources.
- ☐ Amazon GuardDuty
Threat detection service that continuously monitors for malicious activity and unauthorized behavior.

An organization recently faces an attack on their cloud infrastructure. The organization wants to identify the list of AWS Services that are shared externally.

Which AWS Service can the organization use to identify the AWS resources that are shared externally?

- ☐ Amazon OpenSearch Service AWS Managed OpenSearch clusters AI-powered search, observability, and vector database operations (unrelated to resource sharing)
- ☐ AWS Control Tower Automate the setup and governs multi-account AWS environment based on AWS best practices (doesn't detect external sharing).
- ☒ AWS IAM Access Analyzer Helps identify AWS resources shared with external entities. Supports Zone of Trust and Scans S3 buckets, KMS keys, IAM roles, Lambda functions
- ☐ AWS Fargate